



DIGITAL VLSI DESIGN

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DIGITAL VLSI DESIGN

PART 2

Fabrication of MOSFETs with Layout Design

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Fabrication of MOSFETS with Layout Design

UNIT 1 (Part 2): Fabrication of MOSFETS with Layout Design

Introduction, Fabrication Process Flow: Basic Steps (2.1 & 2.2)

The CMOS n-Well Process (2.3)

Layout Design Rules (2.4)

Full-Custom Mask Layout Design (2.5)

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Fabrication of MOSFETs

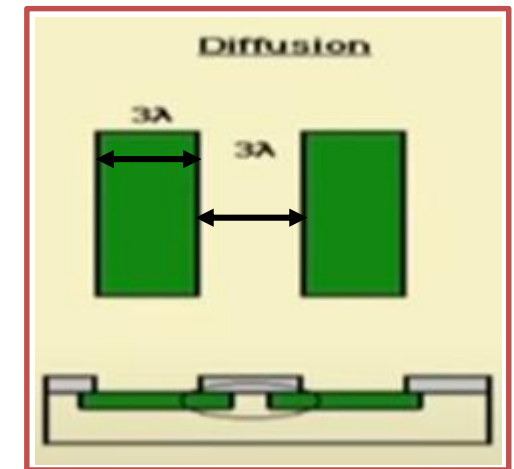
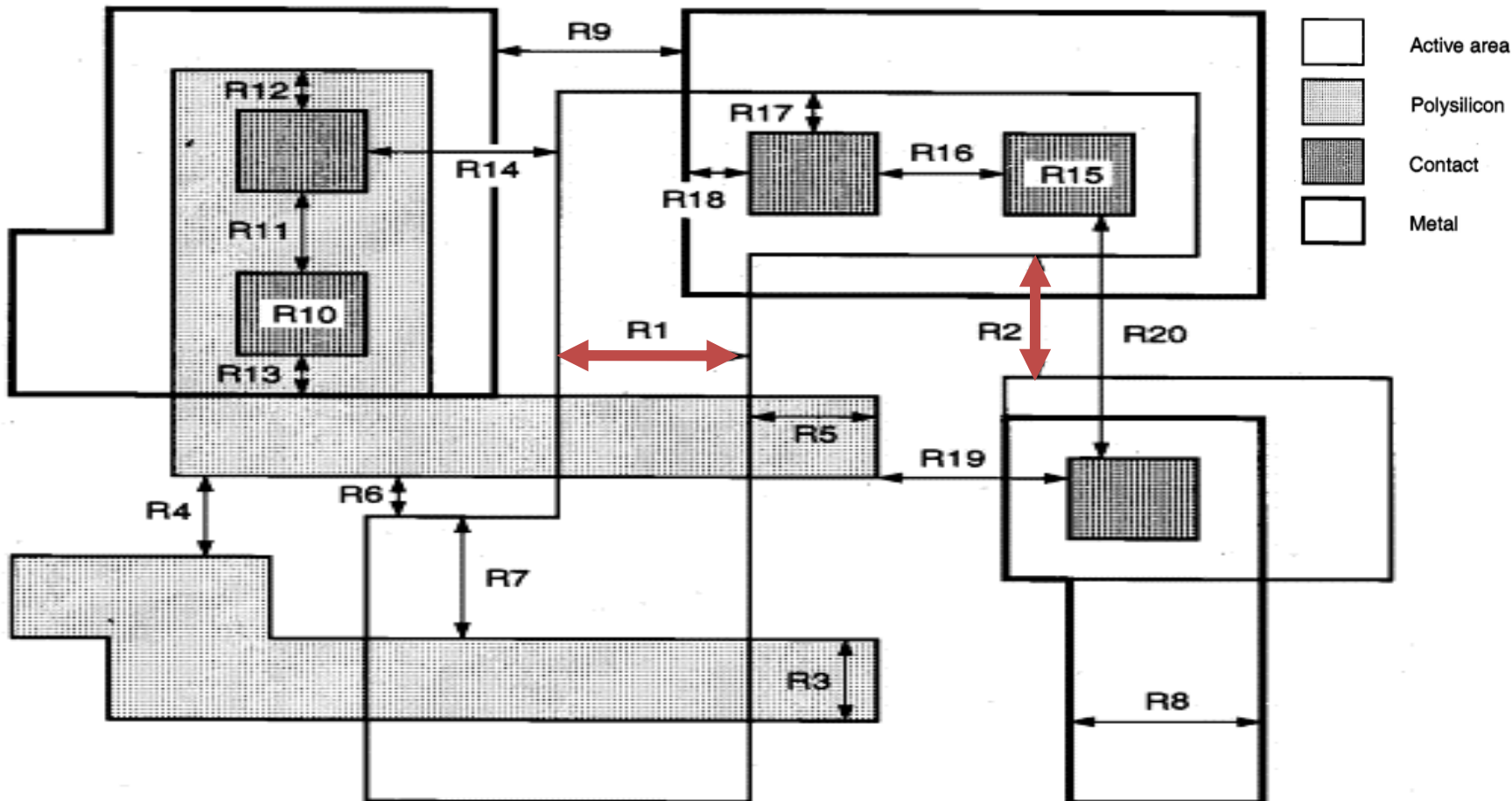
Part c – Layout Design Rules

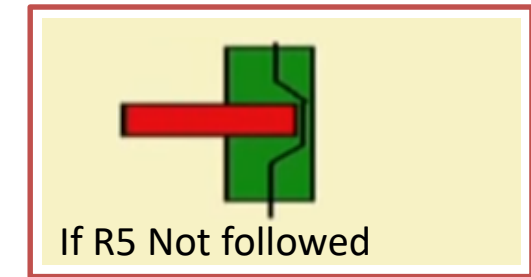
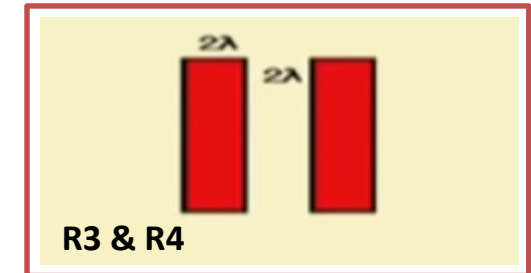
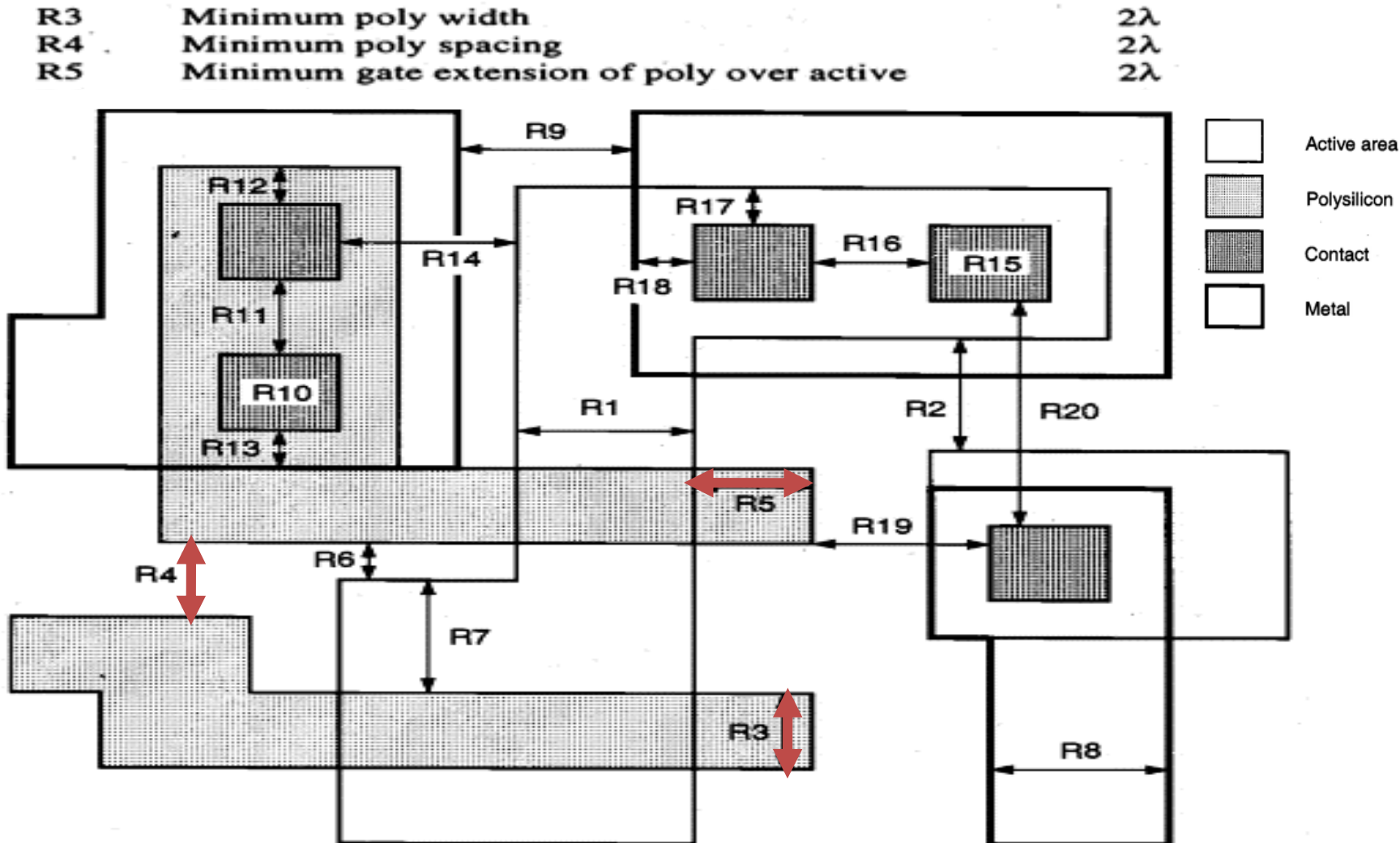
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- In VLSI design, as processes become more complex, it becomes very difficult for the designer to understand the intricacies of the fabrication process and identify errors (if any) in the photo masks.
- **Solution:** Define a set of design rules.
 - Act as interface between the circuit designer and the process engineer during the manufacturing phase.
 - Design rules can be either conservative or aggressive, depending upon whether yield or performance is the main goal.
- Basically, the design rules specify certain geometric constraints on the layout artwork.
 - Ensures that the patterns on the processed wafers will preserve the topology and geometry of the designs.
 - Minimum-width and minimum-spacing constraints.
 - Between objects on the same or different layers.

R1 Minimum active area width 3λ
R2 Minimum active area spacing 3λ

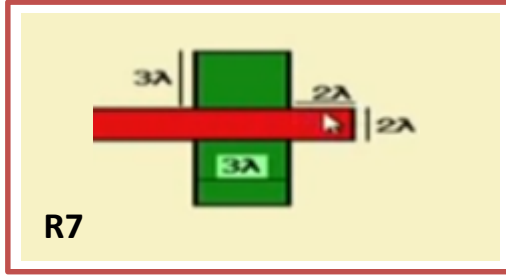
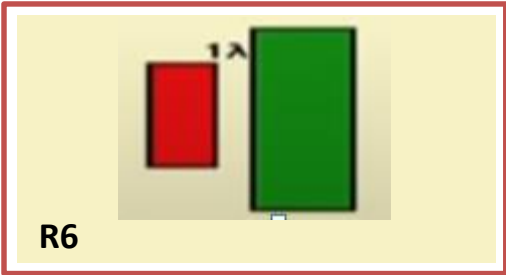
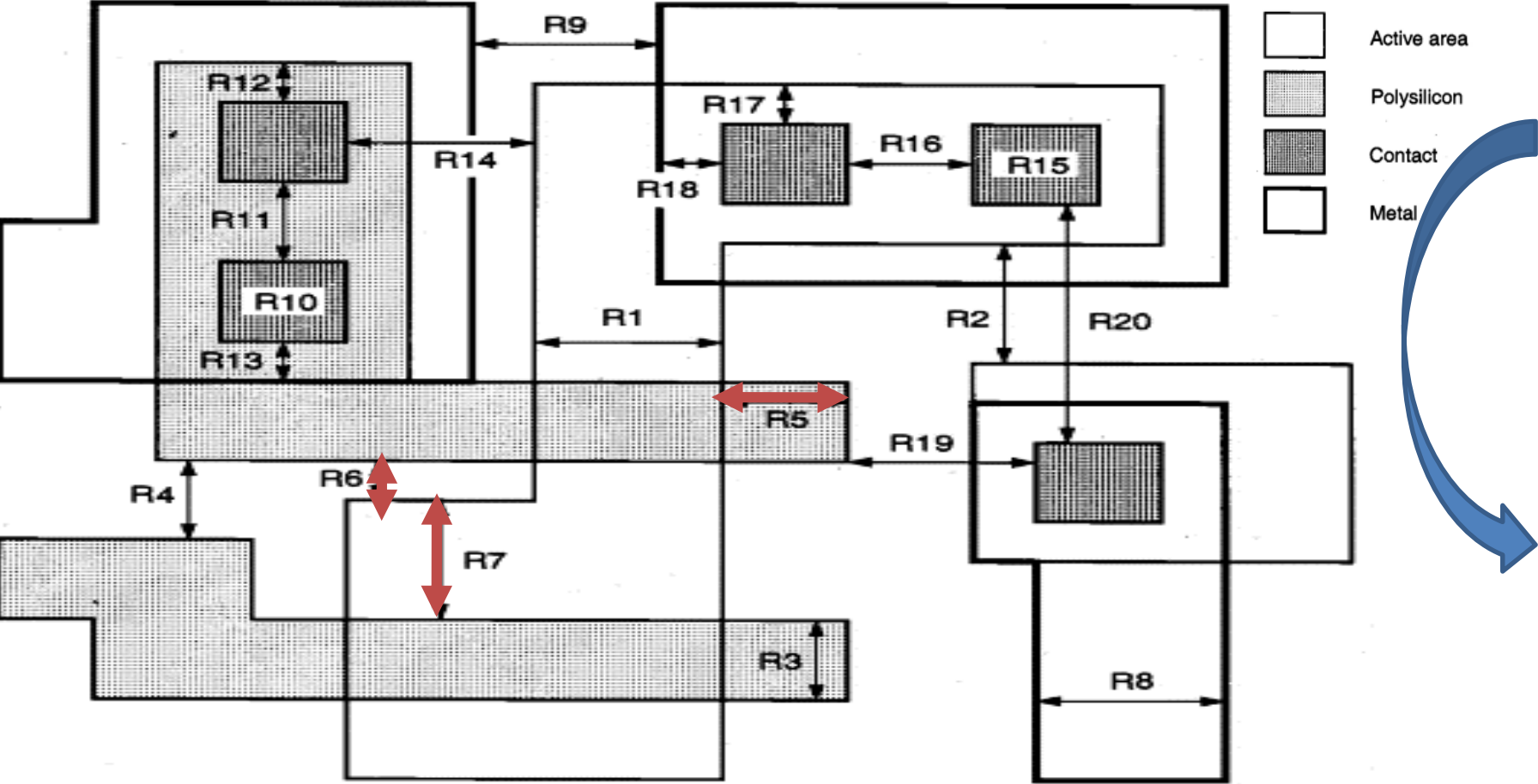


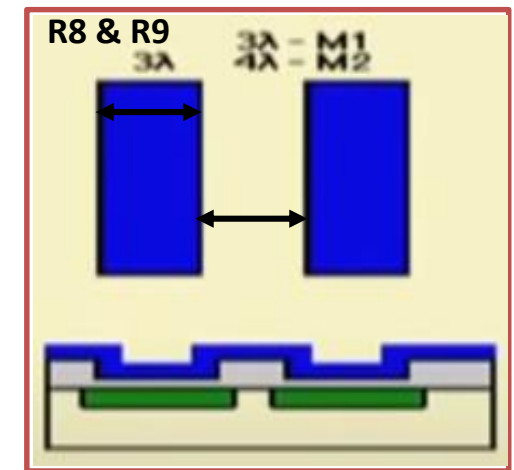
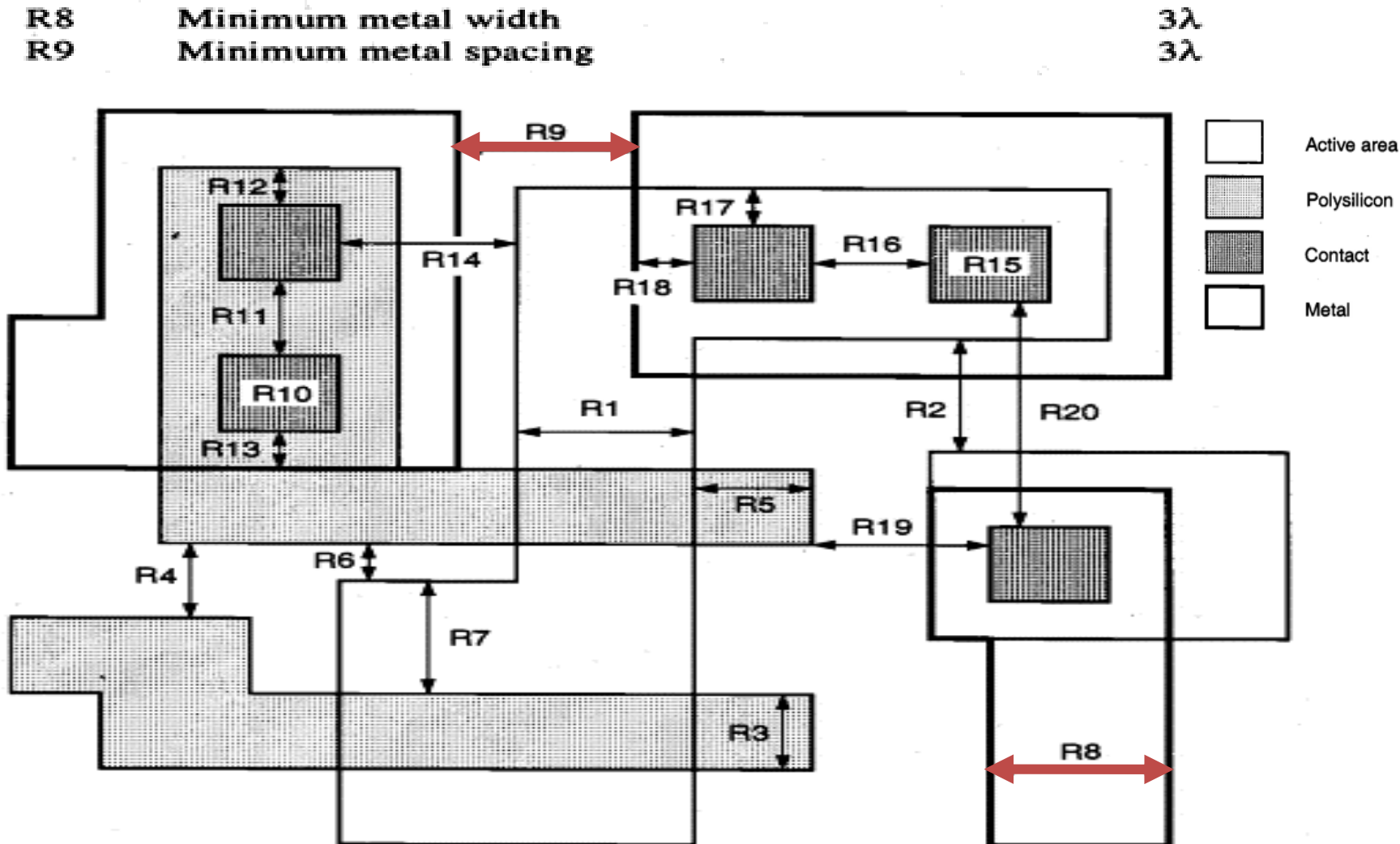


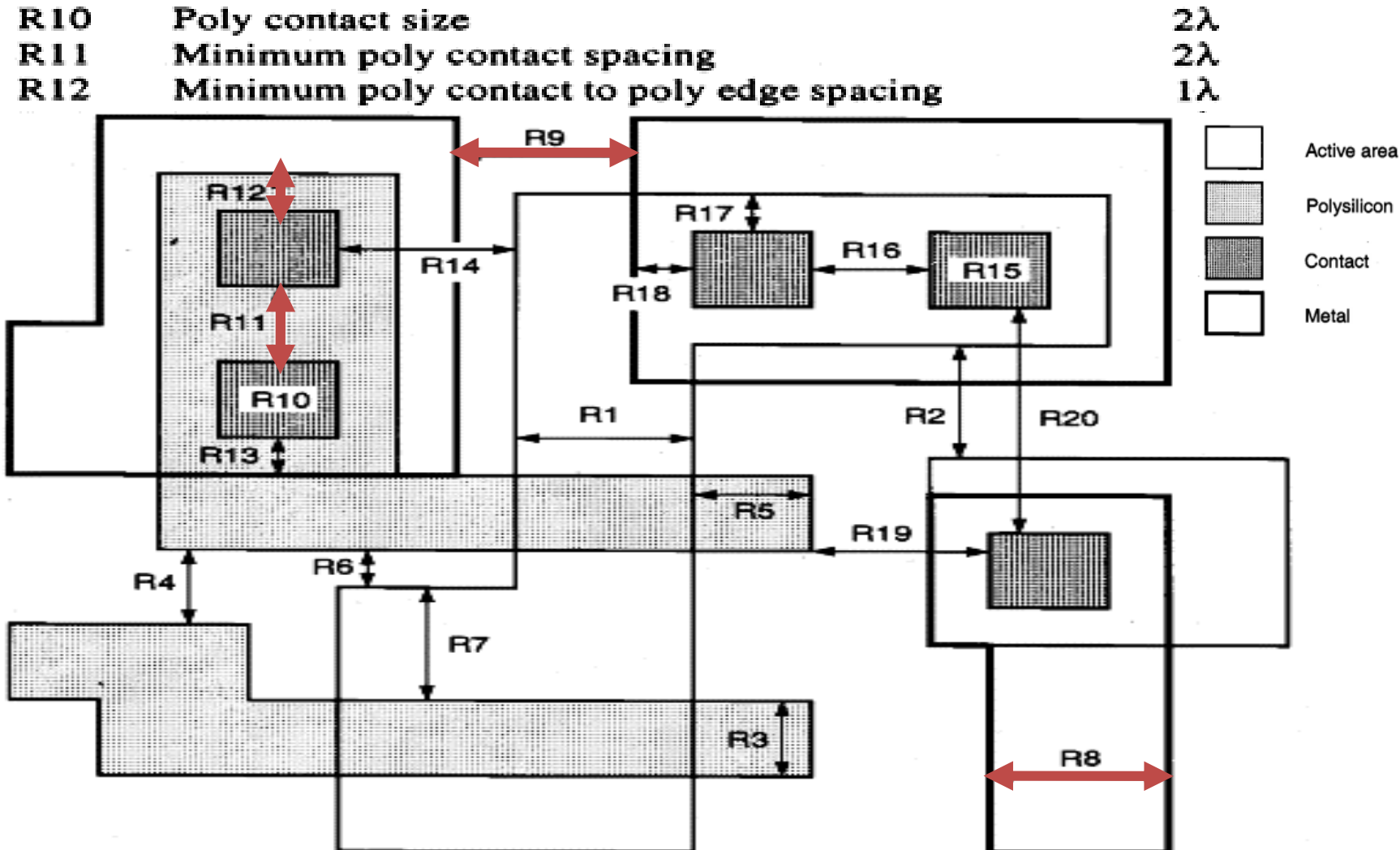
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Lambda based Design Rules – Polysilicon Rules

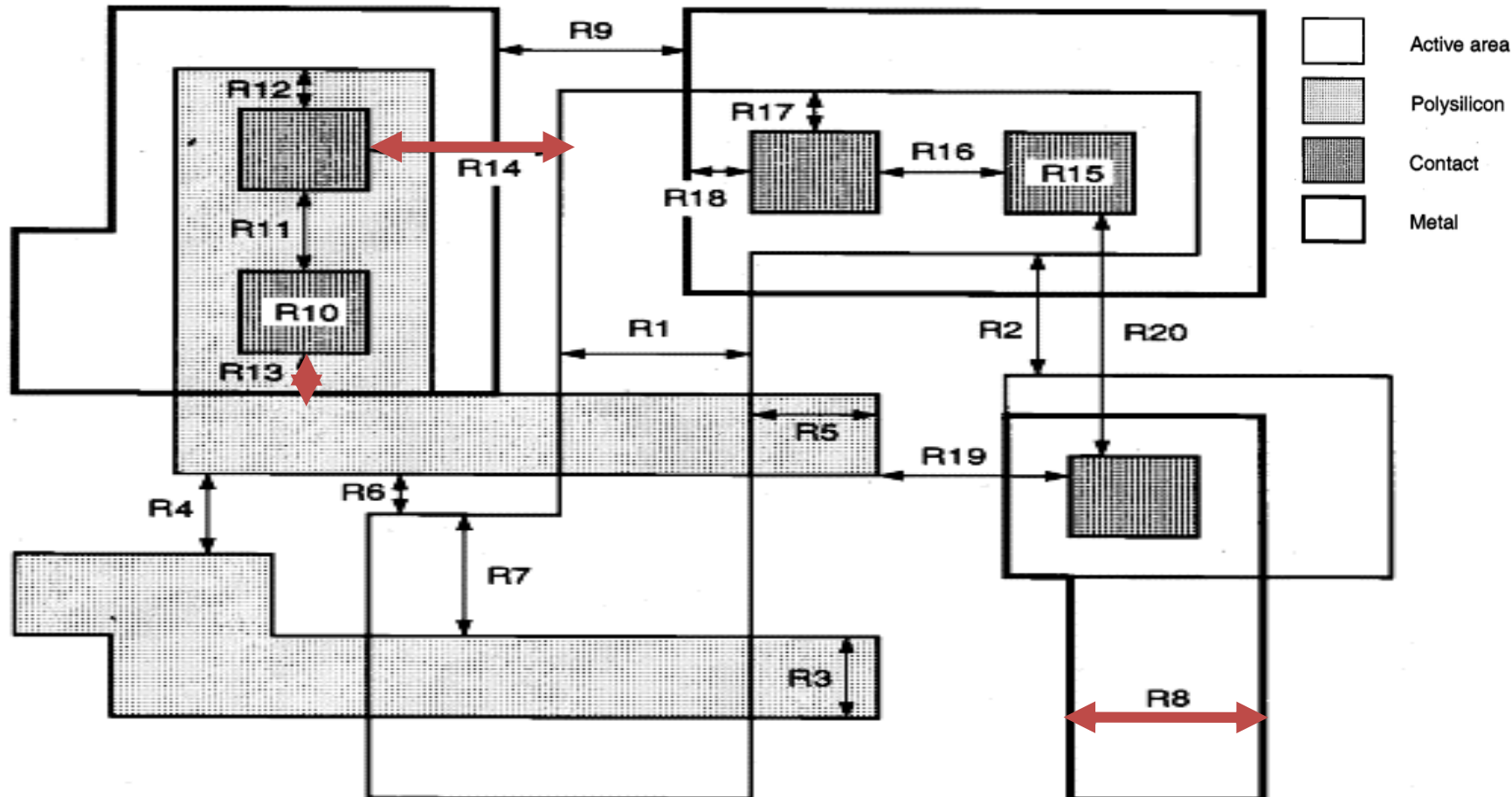
- R6 Minimum poly-active edge spacing (poly outside active area) 1λ
- R7 Minimum poly-active edge spacing (poly inside active area) 3λ

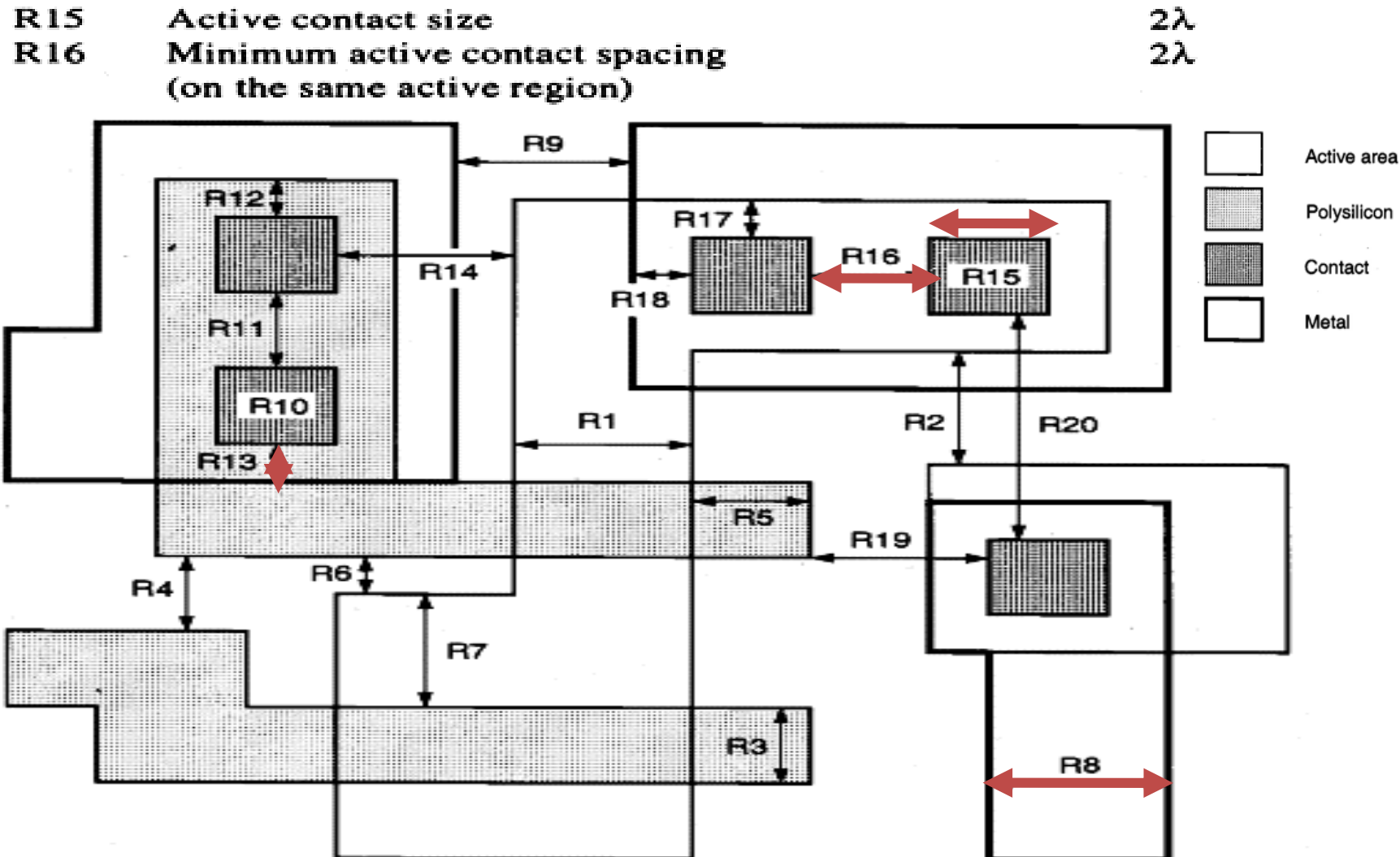




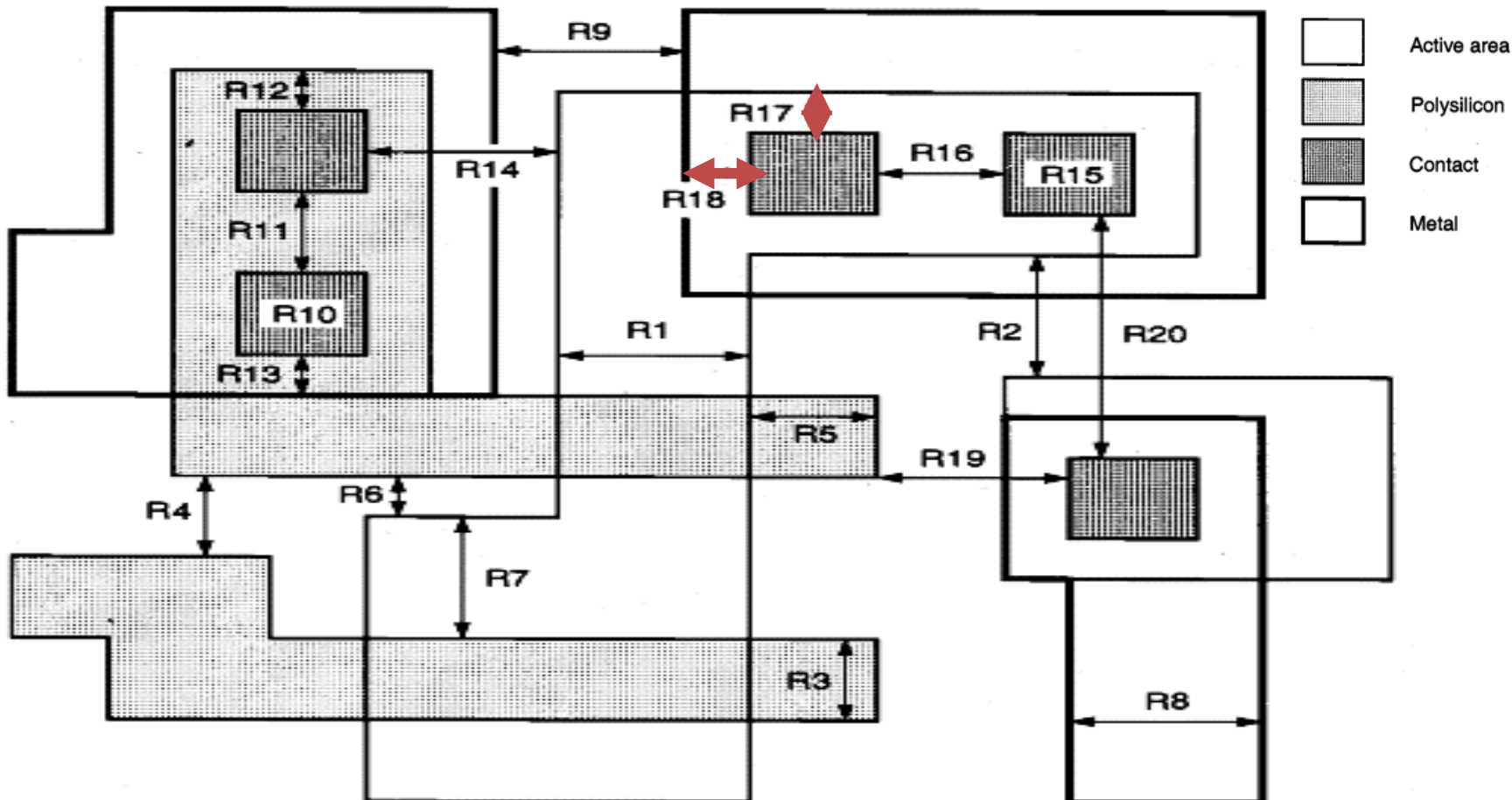


- R13 Minimum poly contact to metal edge spacing 1λ
 R14 Minimum poly contact to active edge spacing 3λ

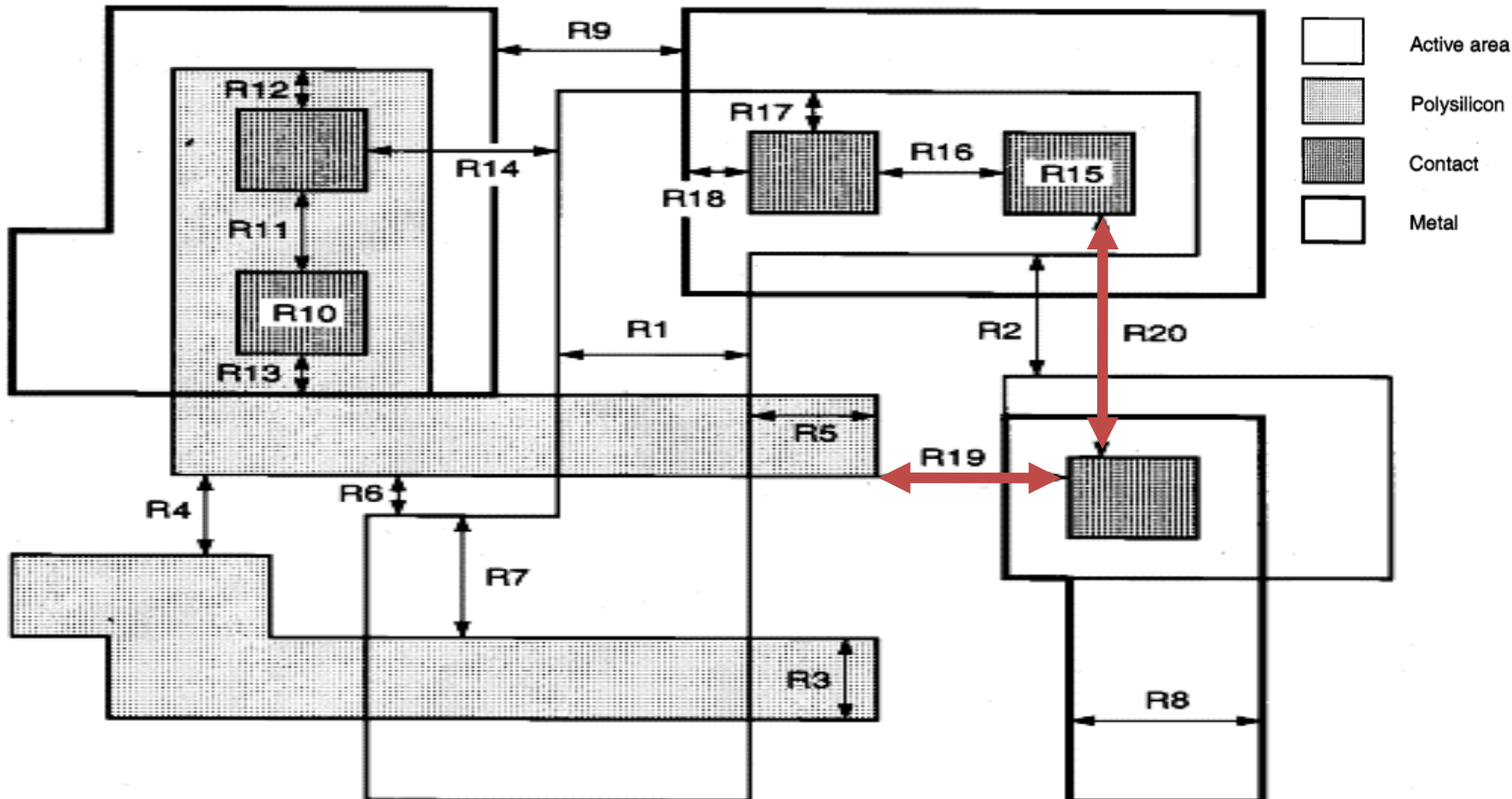




R17	Minimum active contact to active edge spacing	1λ
R18	Minimum active contact to metal edge spacing	1λ

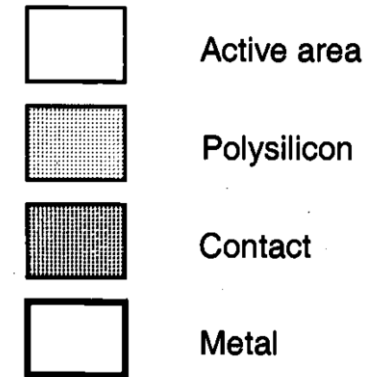
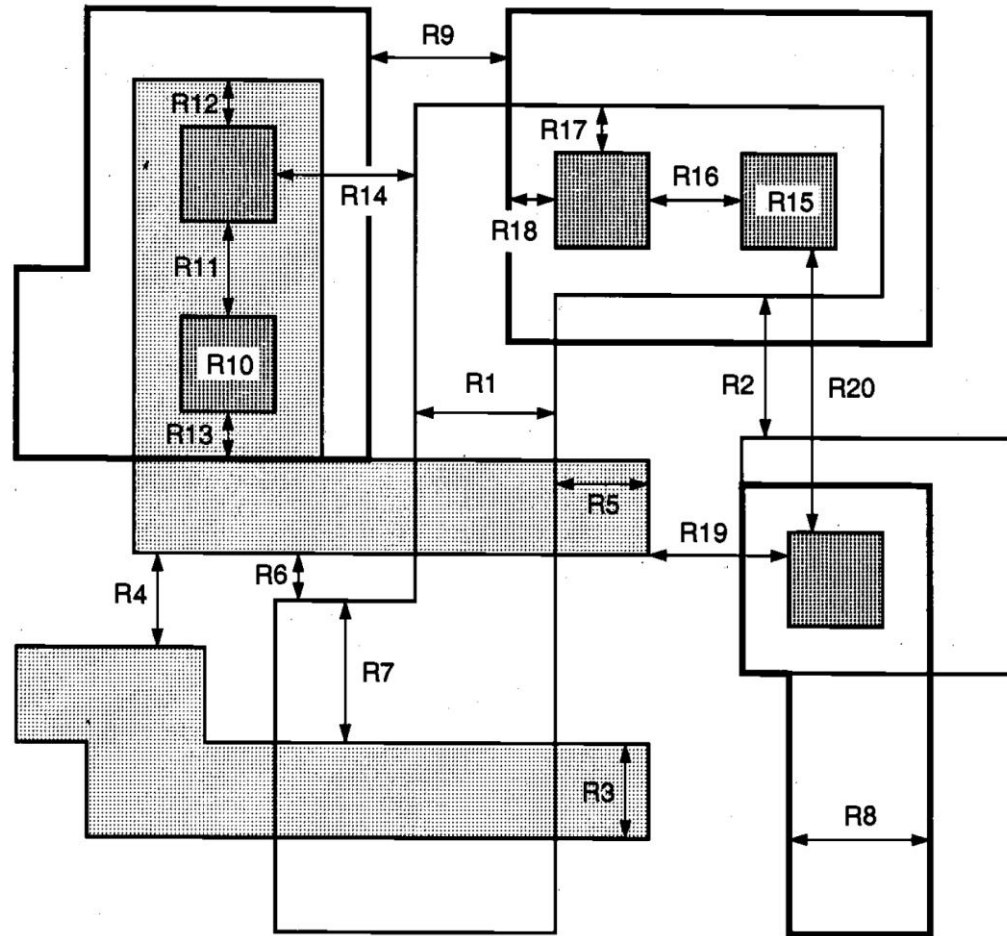


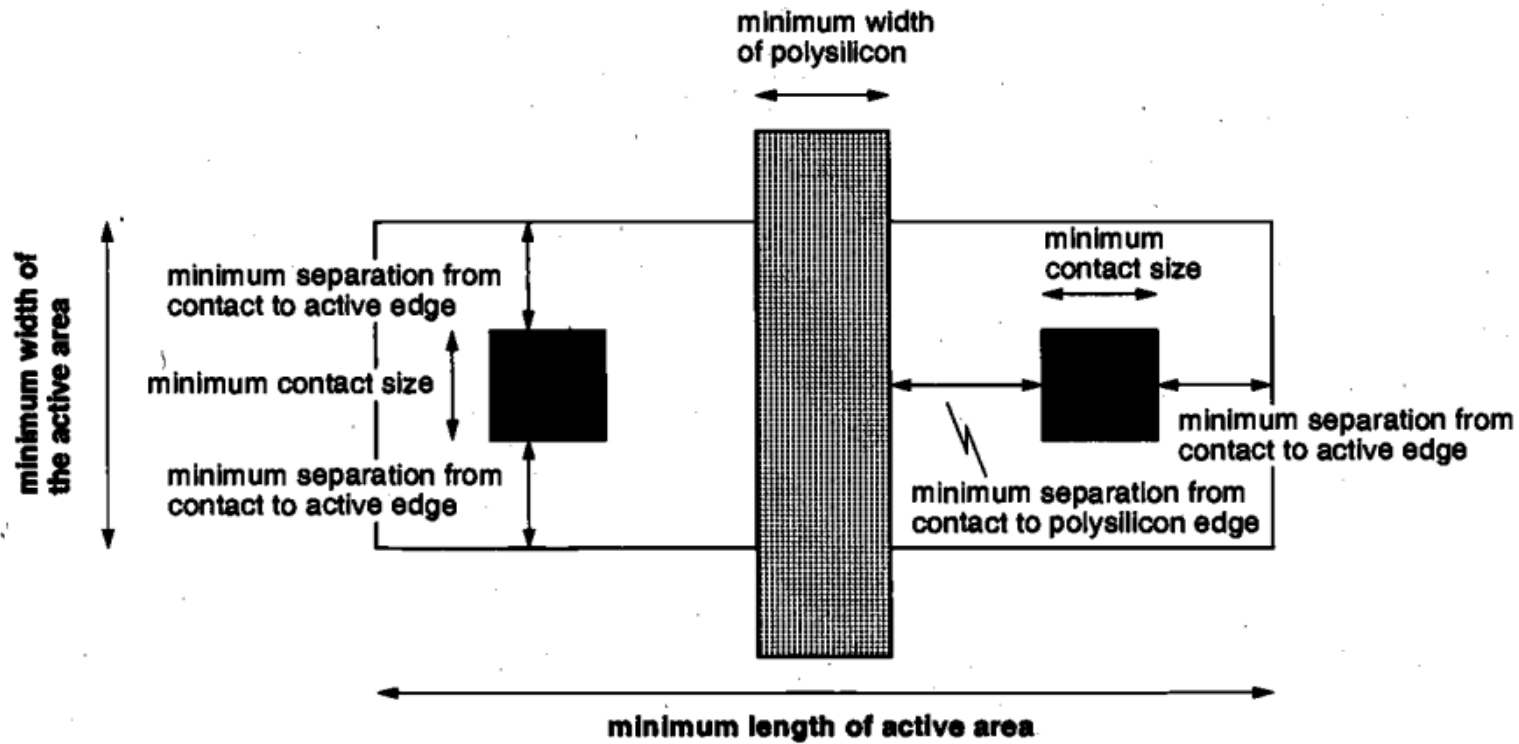
R19	Minimum active contact to poly edge spacing	3λ
R20	Minimum active contact spacing (on different active regions)	6λ

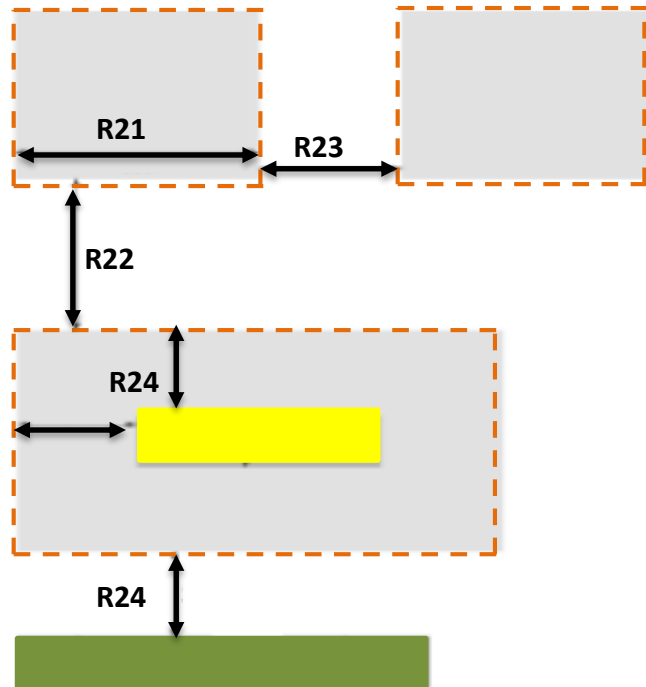


MOSIS Layout Design Rules (sample set)

Rule number	Description	λ -Rule
Active area rules		
R1	Minimum active area width	3λ
R2	Minimum active area spacing	3λ
Polysilicon rules		
R3	Minimum poly width	2λ
R4	Minimum poly spacing	2λ
R5	Minimum gate extension of poly over active	2λ
R6	Minimum poly-active edge spacing (poly outside active area)	1λ
R7	Minimum poly-active edge spacing (poly inside active area)	3λ
Metal rules		
R8	Minimum metal width	3λ
R9	Minimum metal spacing	3λ
Contact rules		
R10	Poly contact size	2λ
R11	Minimum poly contact spacing	2λ
R12	Minimum poly contact to poly edge spacing	1λ
R13	Minimum poly contact to metal edge spacing	1λ
R14	Minimum poly contact to active edge spacing	3λ
R15	Active contact size	2λ
R16	Minimum active contact spacing (on the same active region)	2λ
R17	Minimum active contact to active edge spacing	1λ
R18	Minimum active contact to metal edge spacing	1λ
R19	Minimum active contact to poly edge spacing	3λ
R20	Minimum active contact spacing (on different active regions)	6λ

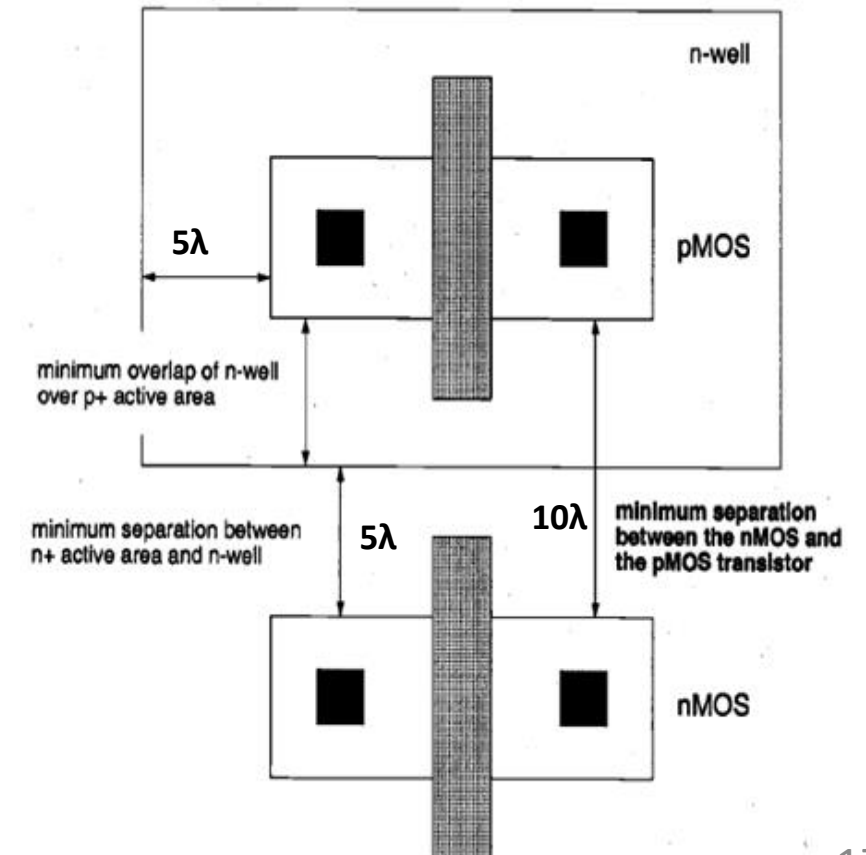


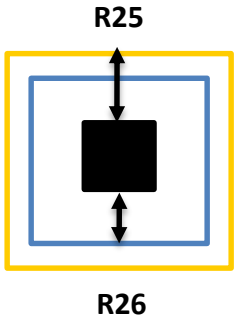
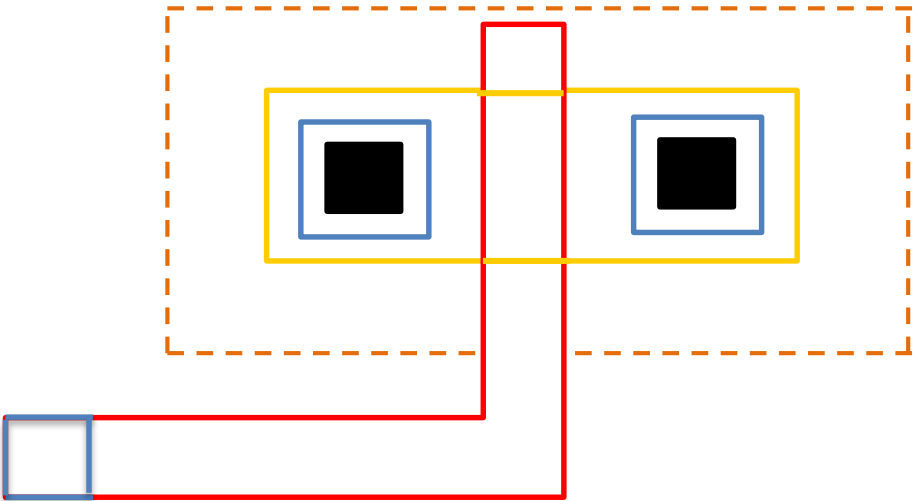




Well (N/P)		
R21	Min width of well	10λ
R22	Minimum Spacing between well at Different potential	9λ
R23	Minimum Spacing between well at same potential	6λ
R24	Active to well edge spacing	5λ

Minimum Separation between NMOS and PMOS





Contact Rules

Contact rules		
R10	Poly contact size	2λ
R11	Minimum poly contact spacing	2λ
R12	Minimum poly contact to poly edge spacing	1λ
R13	Minimum poly contact to metal edge spacing	1λ
R14	Minimum poly contact to active edge spacing	3λ
R15	Active contact size	2λ
R16	Minimum active contact spacing (on the same active region)	2λ
R17	Minimum active contact to active edge spacing	1λ
R18	Minimum active contact to metal edge spacing	1λ
R19	Minimum active contact to poly edge spacing	3λ
R20	Minimum active contact spacing (on different active regions)	6λ

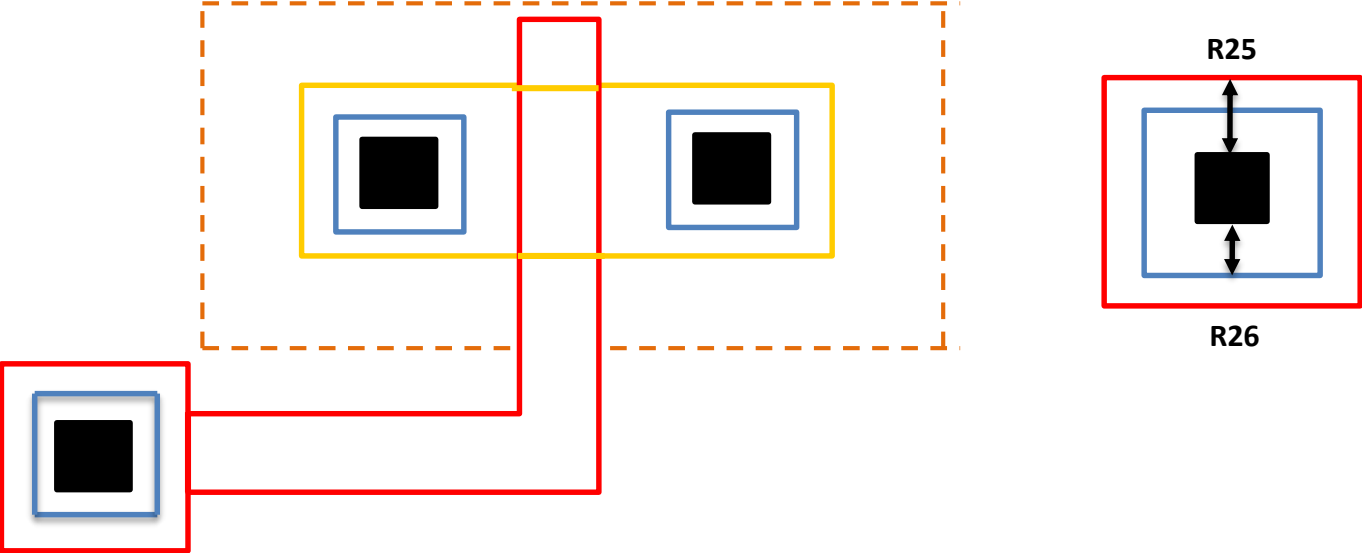
Contact (Additional)		
R25	Minimum Active Overlap over Contact	1.5λ
R26	Minimum Metal1 Overlap over Contact	1λ
R27	Minimum Poly Overlap over Contact	1.5λ

METAL 1 TO POLYSILICON OR TO DIFFUSION

When making connections between **METAL** and either of the other two layers;

The 2λ X 2λ contact cut indicates **an area in which oxide is to be removed down to the underlying polysilicon or diffusion surface.**

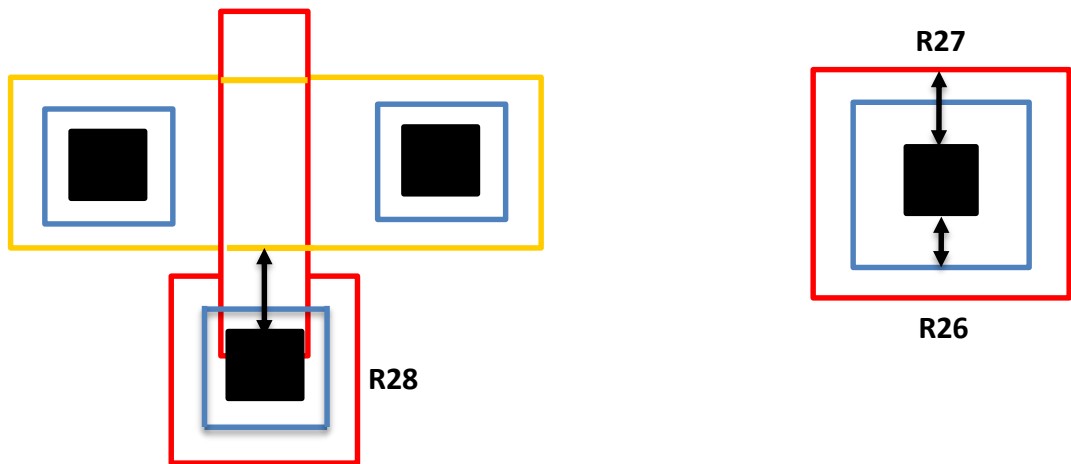
When deposition of the **metal layer takes place the metal is deposited through the contact cut areas onto the underlying area** so that contact is made between the layers



Contact (Additional)		
R25	Minimum Active Overlap over Contact	1.5λ
R26	Minimum Metal1 Overlap over Contact	1λ
R27	Minimum Poly Overlap over Contact	1.5λ
R28	Minimum spacing to gate	2λ

Contact Rules

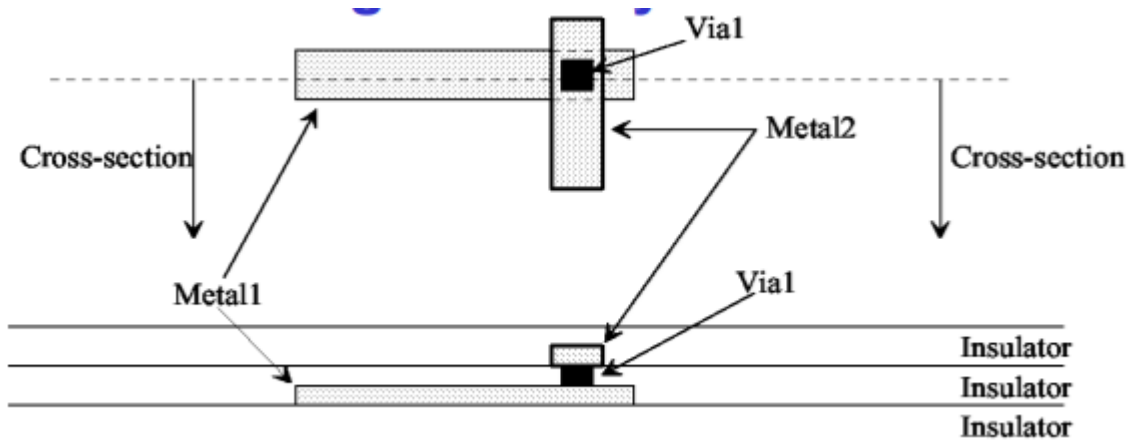
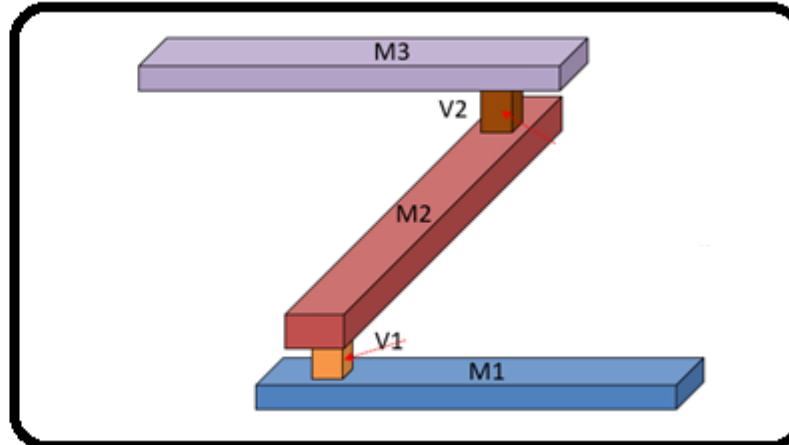
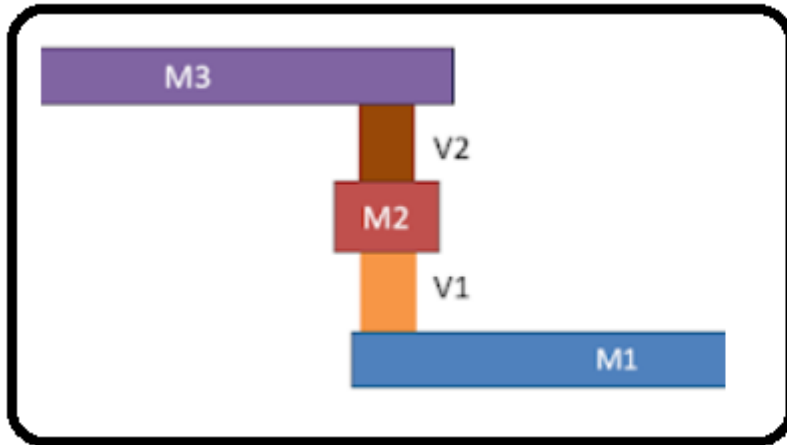
Contact rules		
R10	Poly contact size	2λ
R11	Minimum poly contact spacing	2λ
R12	Minimum poly contact to poly edge spacing	1λ
R13	Minimum poly contact to metal edge spacing	1λ
R14	Minimum poly contact to active edge spacing	3λ
R15	Active contact size	2λ
R16	Minimum active contact spacing (on the same active region)	2λ
R17	Minimum active contact to active edge spacing	1λ
R18	Minimum active contact to metal edge spacing	1λ
R19	Minimum active contact to poly edge spacing	3λ
R20	Minimum active contact spacing (on different active regions)	6λ

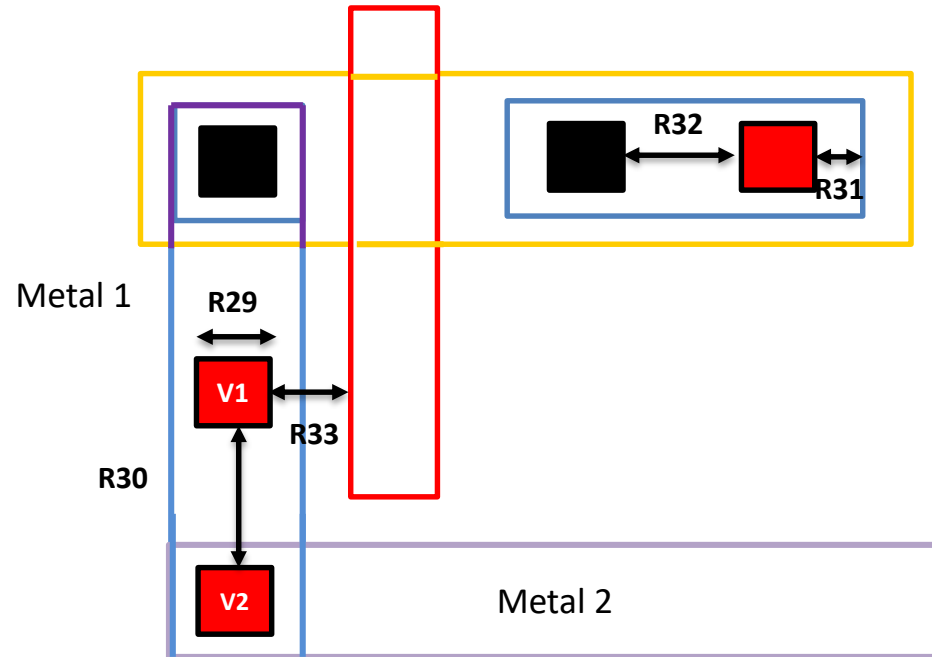


Contact (Additional)		
R25	Minimum Active Overlap over Contact	1.5λ
R26	Minimum Metal1 Overlap over Contact	1λ
R27	Minimum Poly Overlap over Contact	1.5λ
R28	Minimum spacing to gate	2λ

Contact Rules

Contact rules		
R10	Poly contact size	2λ
R11	Minimum poly contact spacing	2λ
R12	Minimum poly contact to poly edge spacing	1λ
R13	Minimum poly contact to metal edge spacing	1λ
R14	Minimum poly contact to active edge spacing	3λ
R15	Active contact size	2λ
R16	Minimum active contact spacing (on the same active region)	2λ
R17	Minimum active contact to active edge spacing	1λ
R18	Minimum active contact to metal edge spacing	1λ
R19	Minimum active contact to poly edge spacing	3λ
R20	Minimum active contact spacing (on different active regions)	6λ

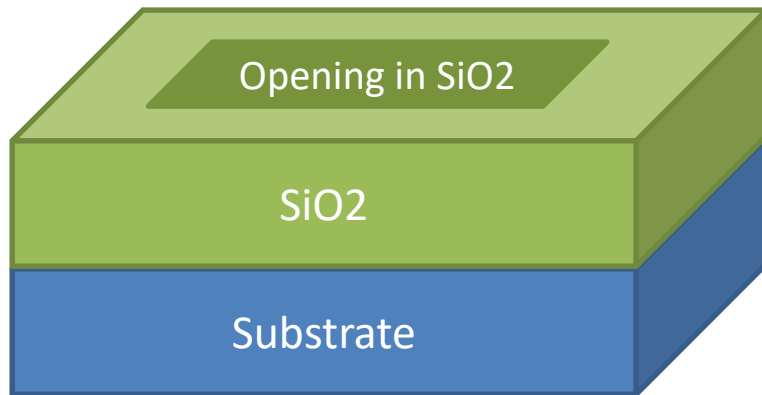




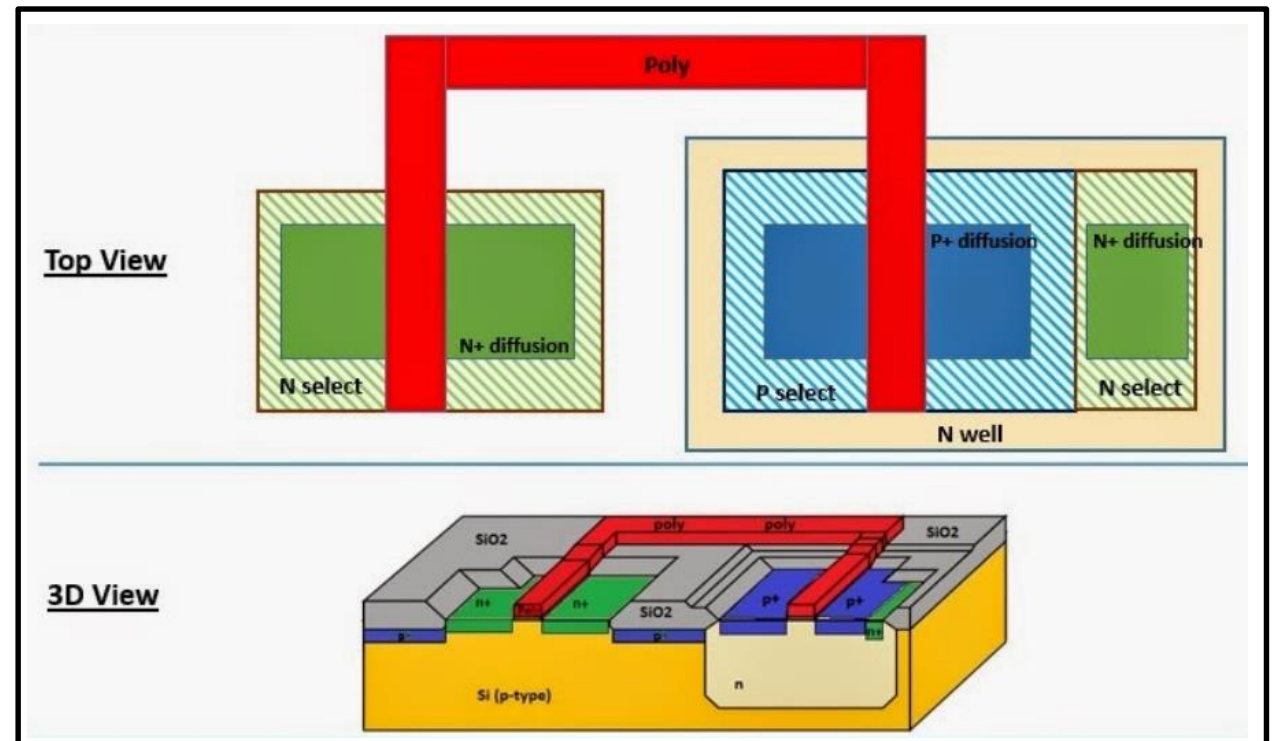
Via Rules		
R29	Exact Size	2λ
R30	Minimum Spacing	3λ
R31	Minimum Overlap by Meta 1	1λ
R32	Minimum spacing to contact	2λ
R33	Minimum Spacing to poly or Active edge	2λ

In Layout

Active Layer : It defines the openings in the SiO₂ covering the substrate.



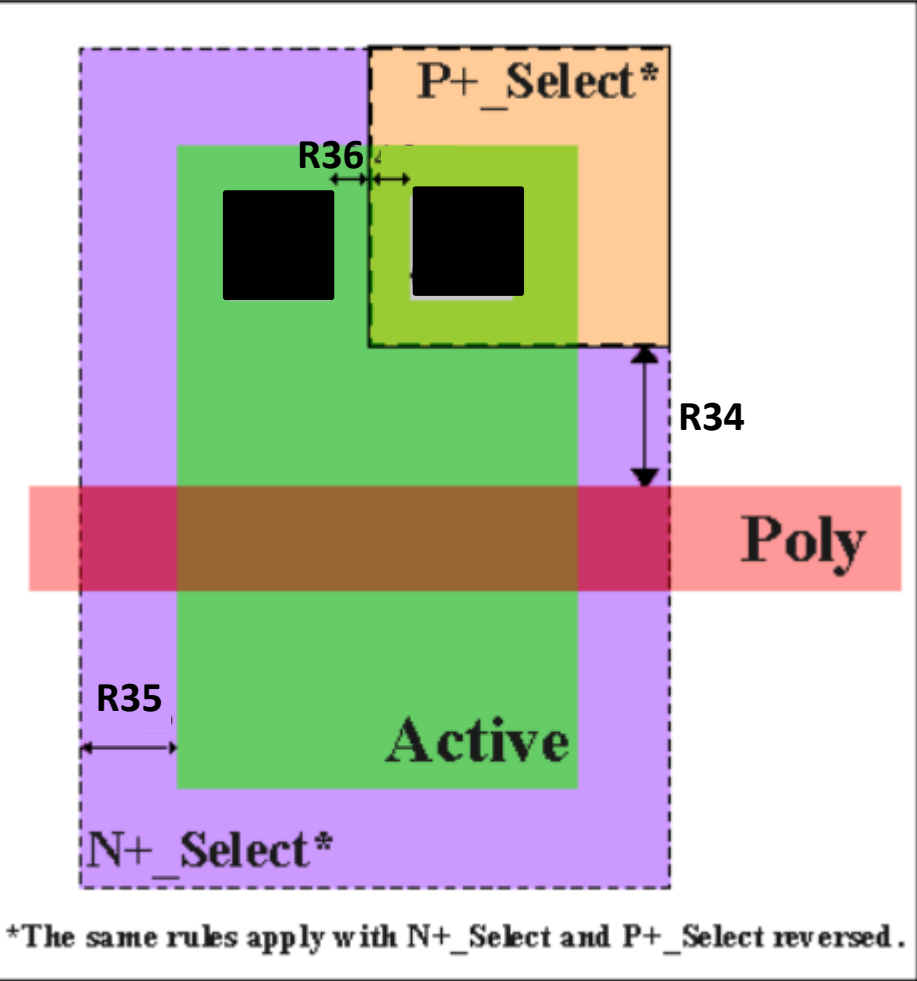
- N/P Select Layer : It defines area (in Active Region) where the n or p type atoms to be implanted
- If it N Select – n type of atoms are implanted and If it P Select – p type of atoms are implanted



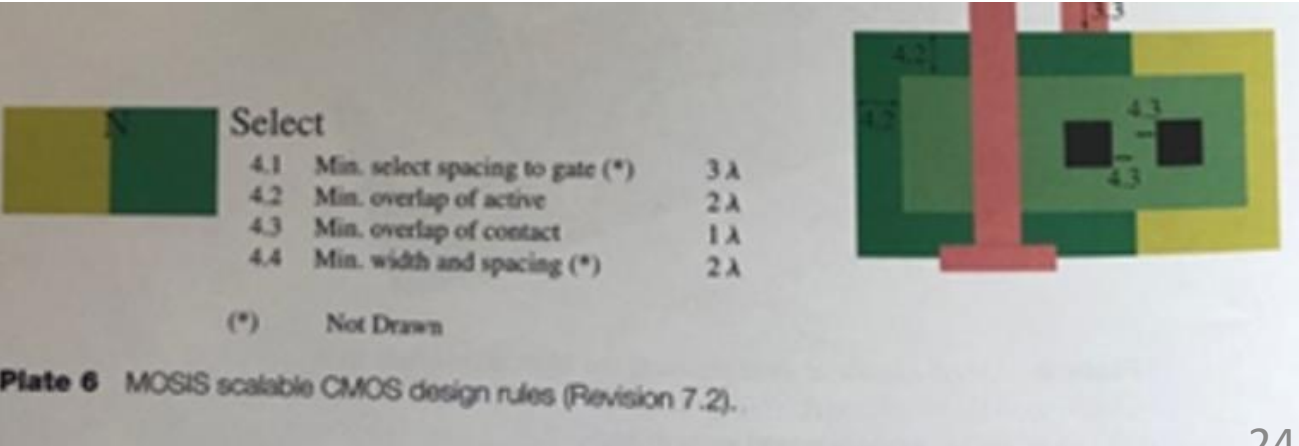
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Lambda based Design Rules – Via Rules

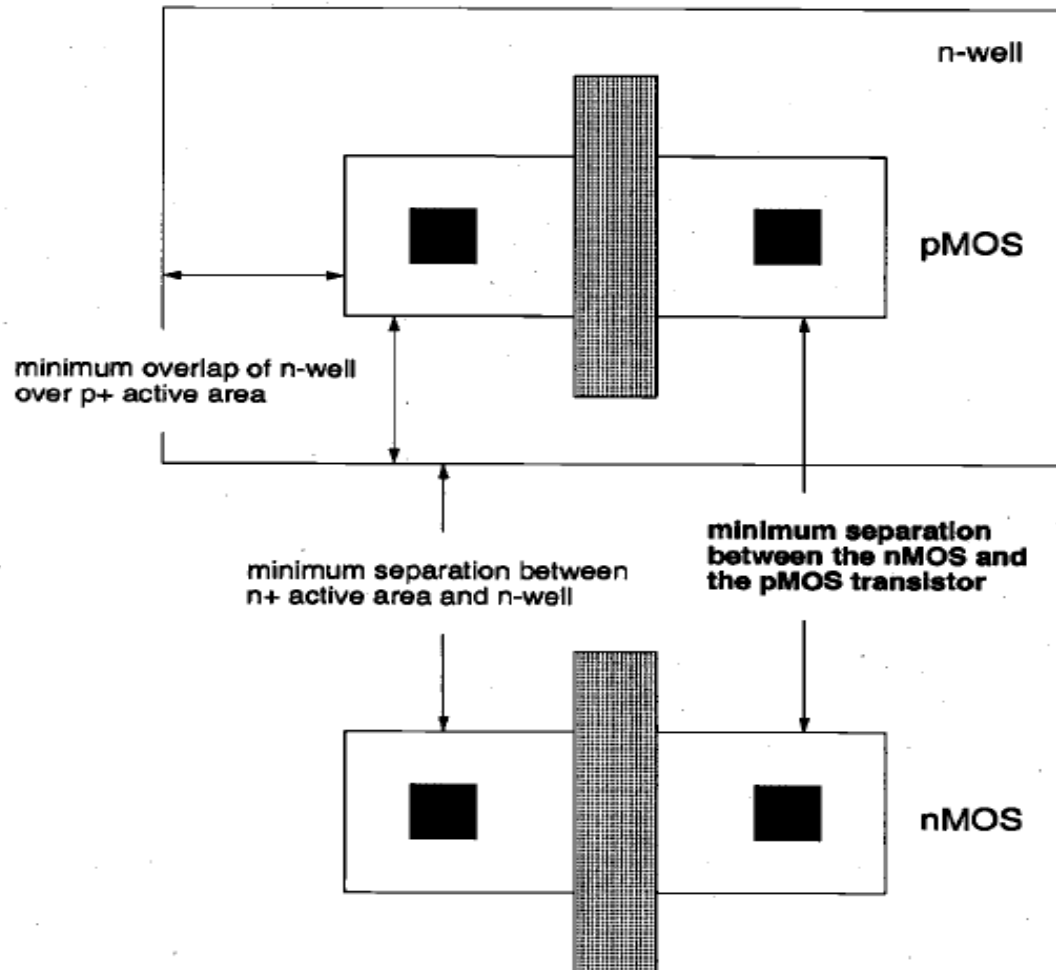
N/P Select Rules



N/P Select Layers		
R34	Minimum Select Spacing to Gate	3λ
R35	Minimum Overlap of Active	2λ
R36	Minimum Overlap of Contact	1λ
R37	Minimum width and Spacing	2λ

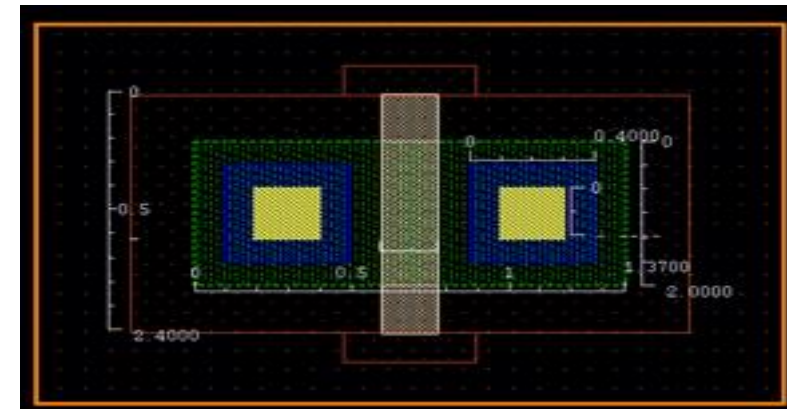


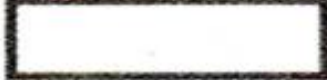
Design rules which determine the separation between the nMOS and the pMOS transistor of the CMOS inverter.



n-well:

1. Minimum overlap of n-well over p+ active area is 5 lambda ($0.5\mu\text{m}$ in 180nm technology)
2. Minimum spacing between two n-well is 5 lambda ($0.5\mu\text{m}$ in 180nm technology).
3. Minimum spacing between n+ active and n-well is 5 lambda ($0.5\mu\text{m}$ in 180nm technology)



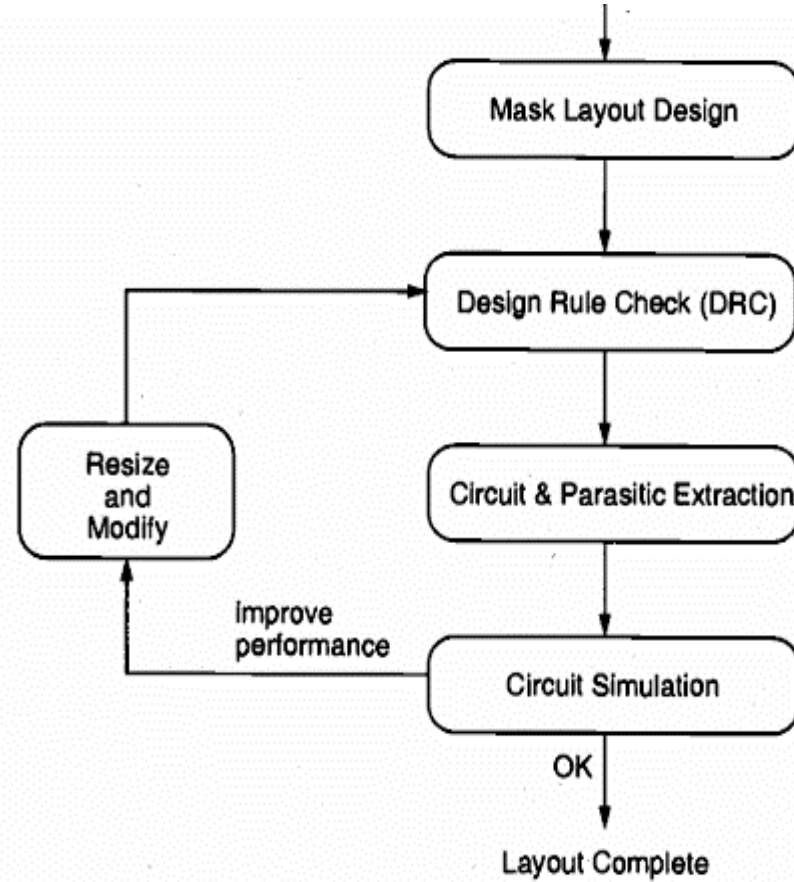
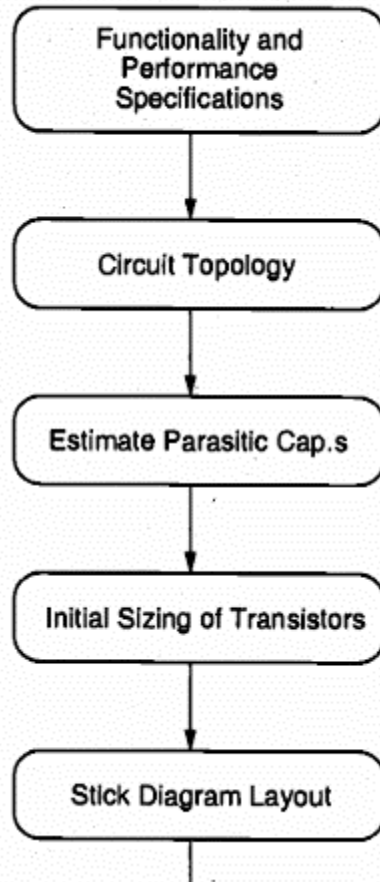
COLOR	STICK ENCODING	LAYERS	MASK LAYOUT ENCODING
GREEN		[n-diffusion (n ⁺ active) Thinox*]	 *Thinox = n-diff. + transistor channels
RED		Polysilicon	
BLUE		Metal 1	
BLACK		Contact cut	
GRAY	NOT APPLICABLE	Overglass	
nMOS ONLY YELLOW		Implant	
nMOS ONLY BROWN		Buried contact	

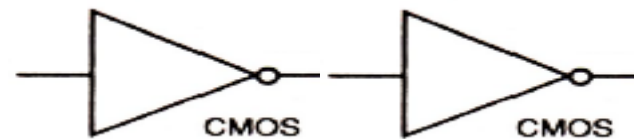
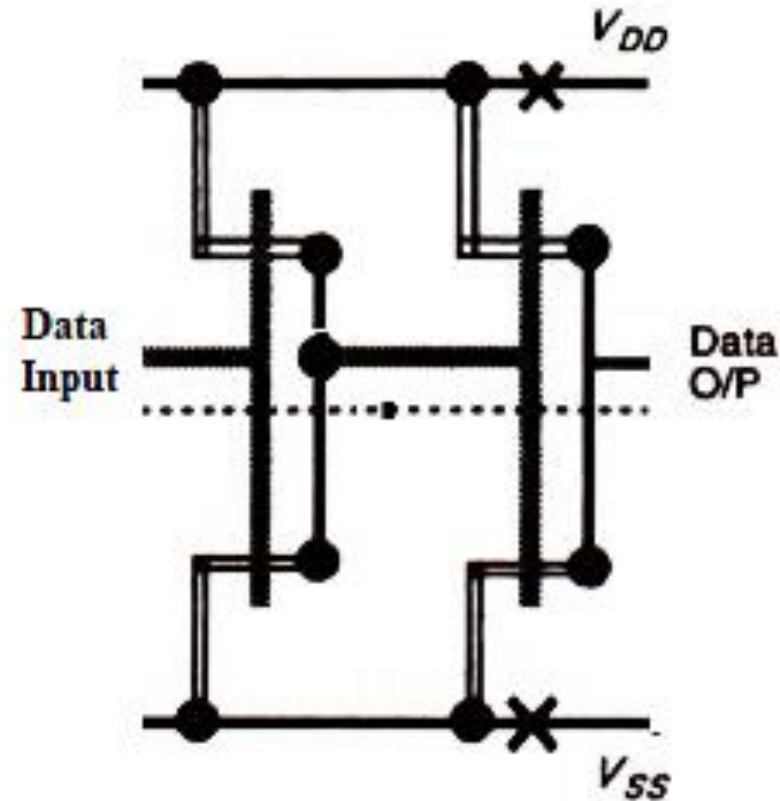
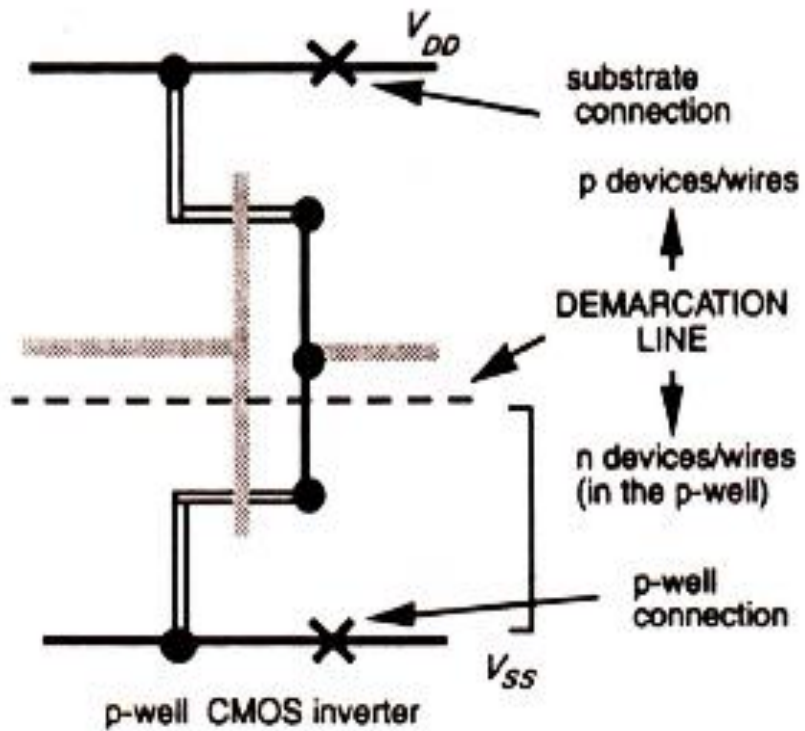
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Encoding form Simple Single Metal NMOS process

COLOR	STICK ENCODING	LAYERS	MASK LAYOUT ENCODING
GREEN		[n-diffusion (n+ active) Thinox*]	
RED		Polysilicon	
BLUE		Metal 1	
BLACK		Contact cut	
GRAY	NOT APPLICABLE	Overglass	
nMOS ONLY YELLOW		Implant	
nMOS ONLY BROWN		Buried contact	

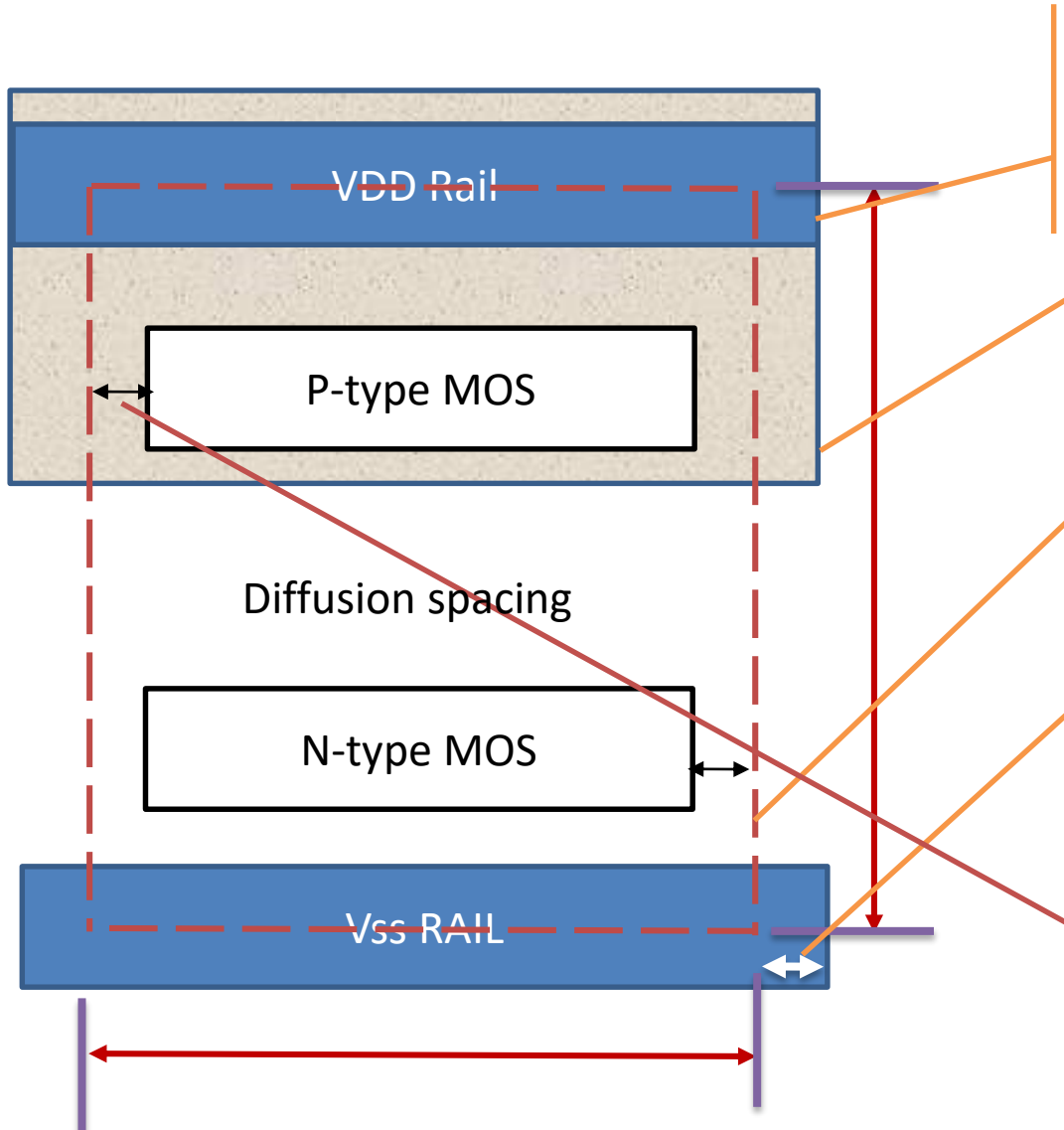
FEATURE	FEATURE (STICK)	FEATURE (SYMBOL)	FEATURE (MASK)
n-type enhancement mode transistor			
Transistor length to width ratio L: W should be shown.			
n-type depletion mode transistor nMOS only			
Source, drain and gate labelling will not normally be shown.			





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Standard Cells Template with Height



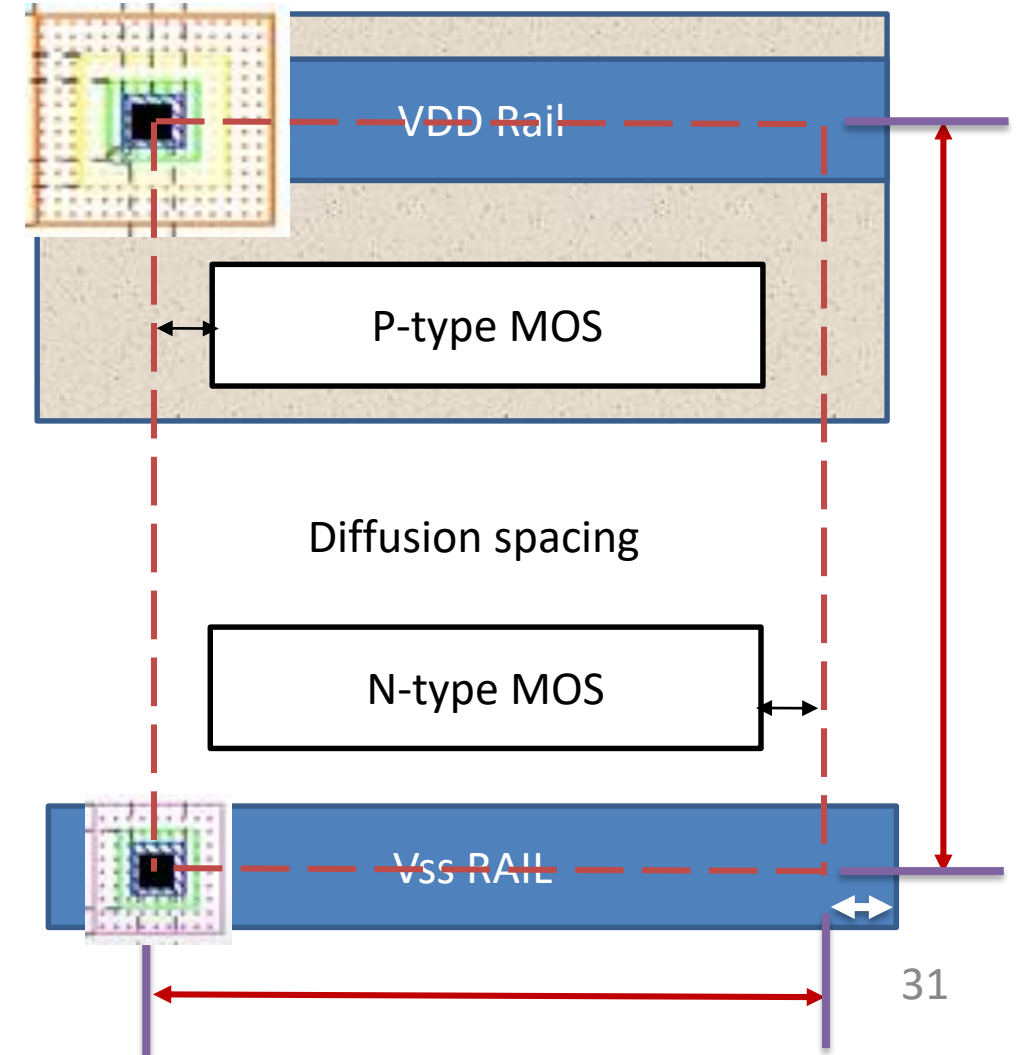
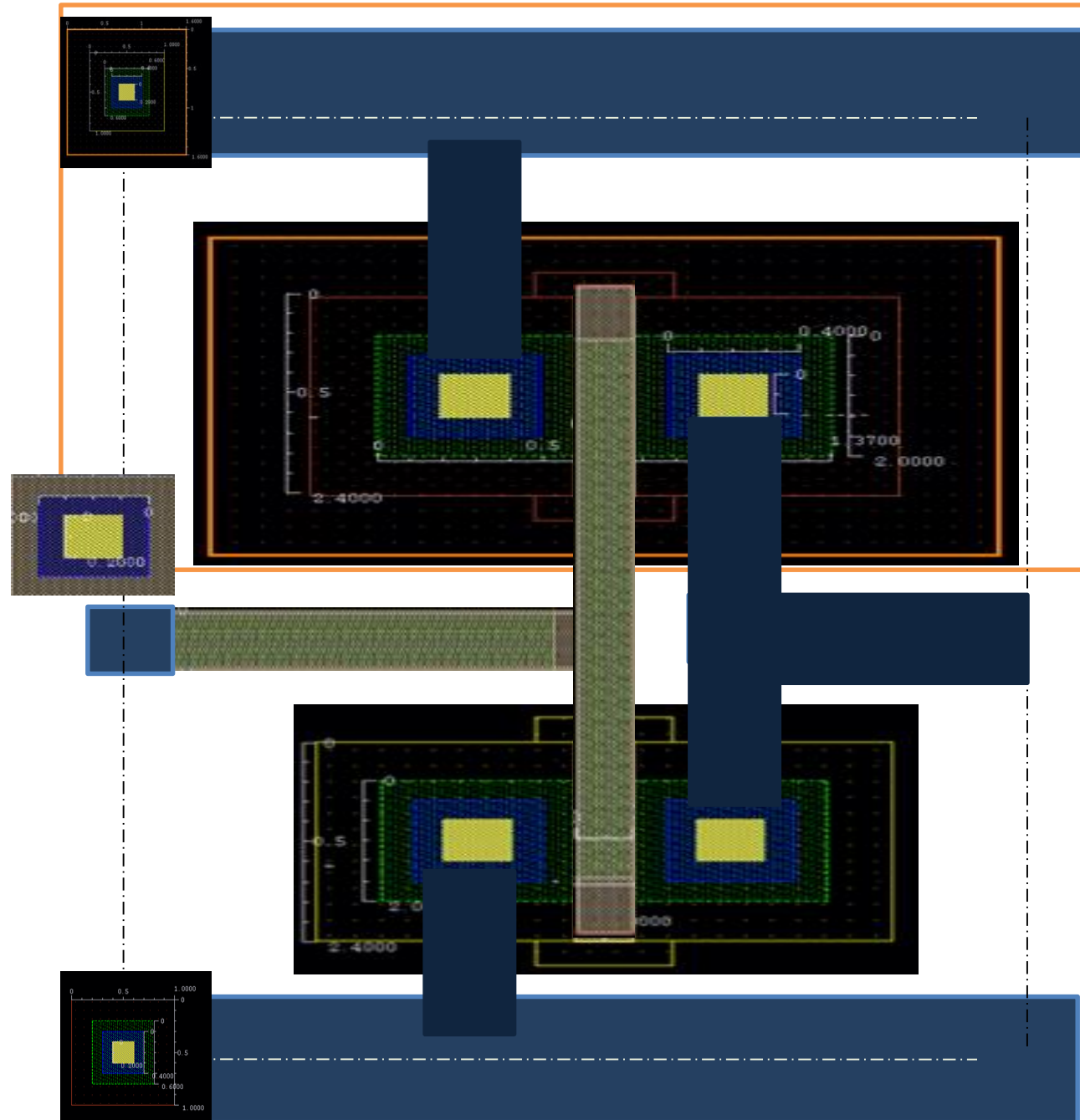
VDD Rail – Generally (3-5)times the Minimum width of the Metal1. Thick Vdd and Gnd lines are used to provide adequate current to ALL the cells in the ROW

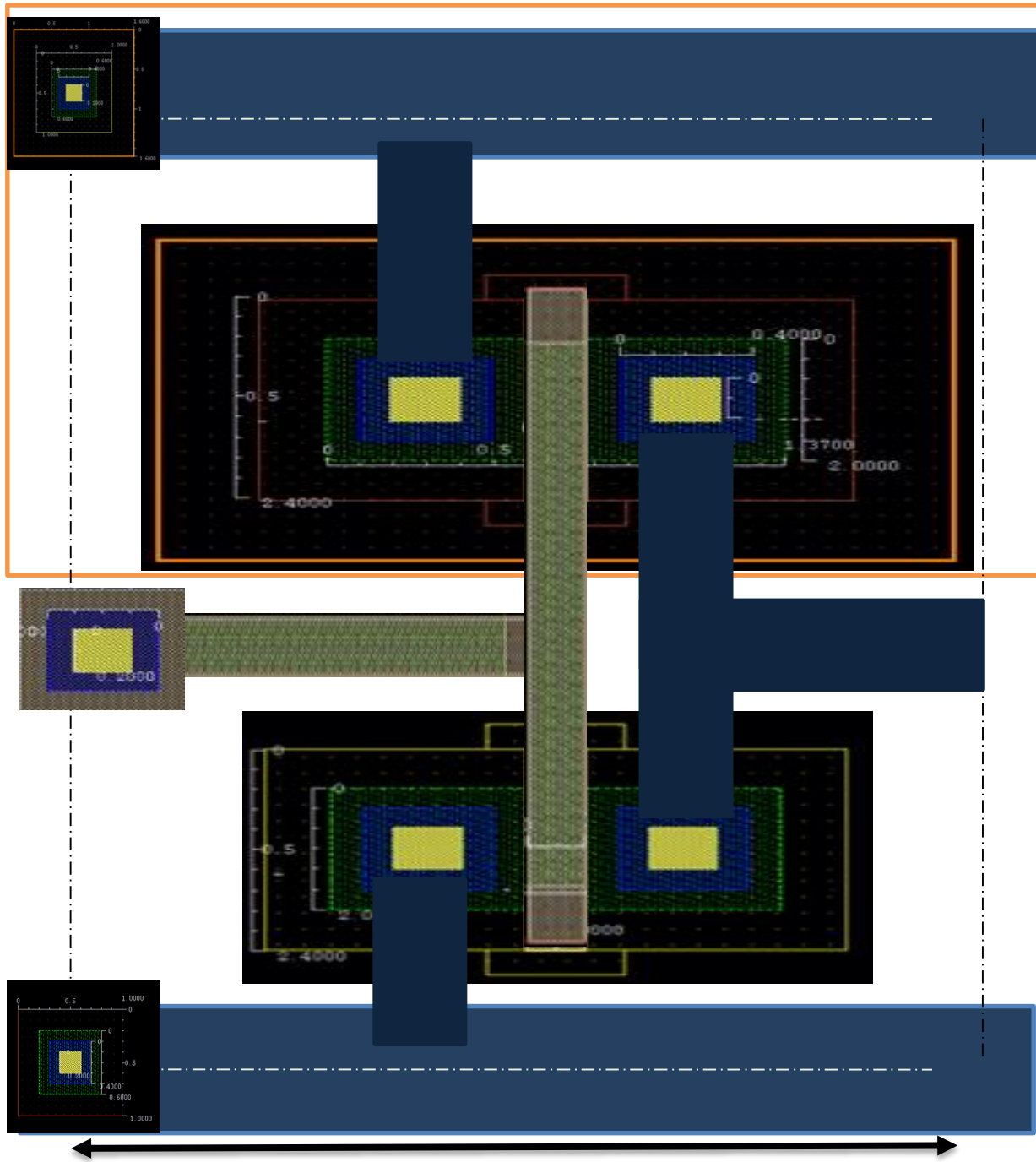
Outside Place and Route (PR) boundary, Nwell is extended so that N-well of neighboring PMOS cells merges – i.e., Common N-Well

The PR boundary : This gives size of cell- It's height is taken from center of VDD and GND line and width by excluding the extensions

Outside PR boundary, The VDD and GND lines are extended which helps to abut the neighboring cells automatically

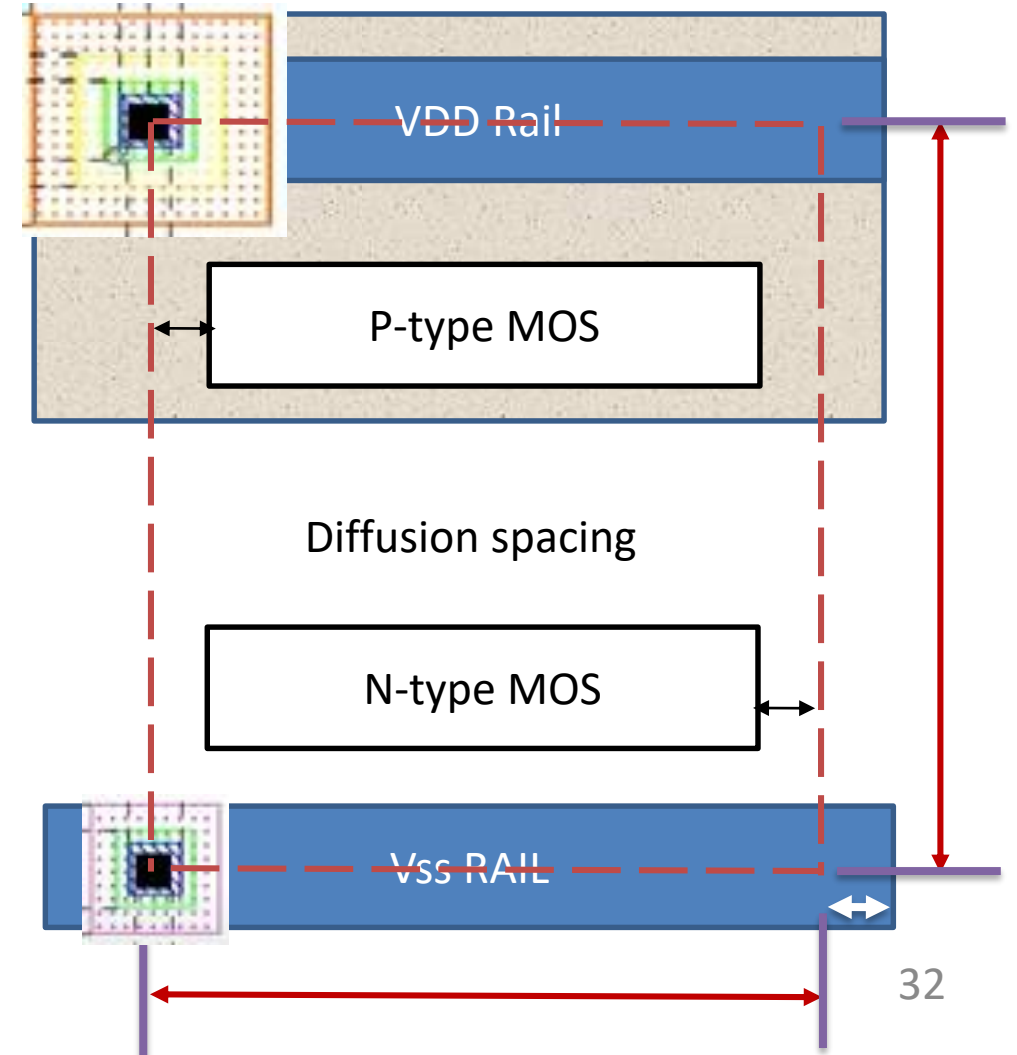
Spacing between PR boundary and n/p type active region is to provide spacing/isolation between neighboring standard cell MOS devices





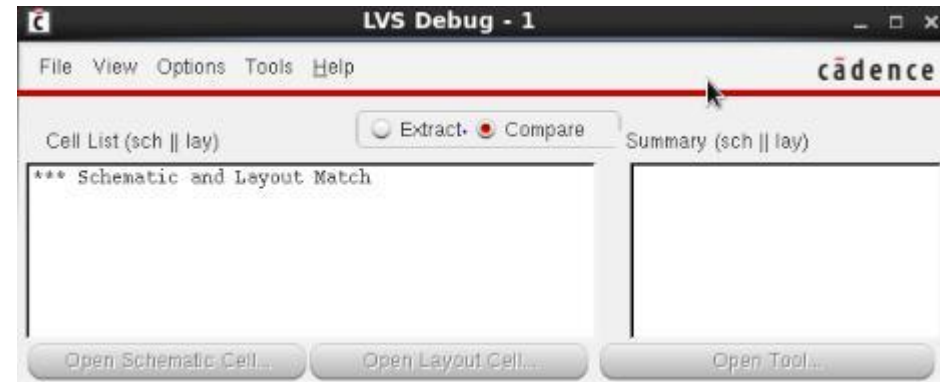
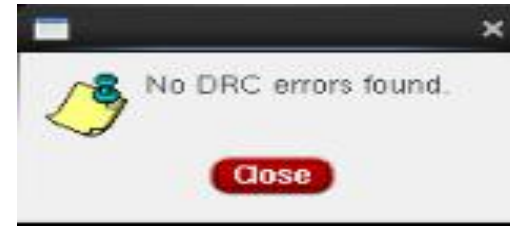
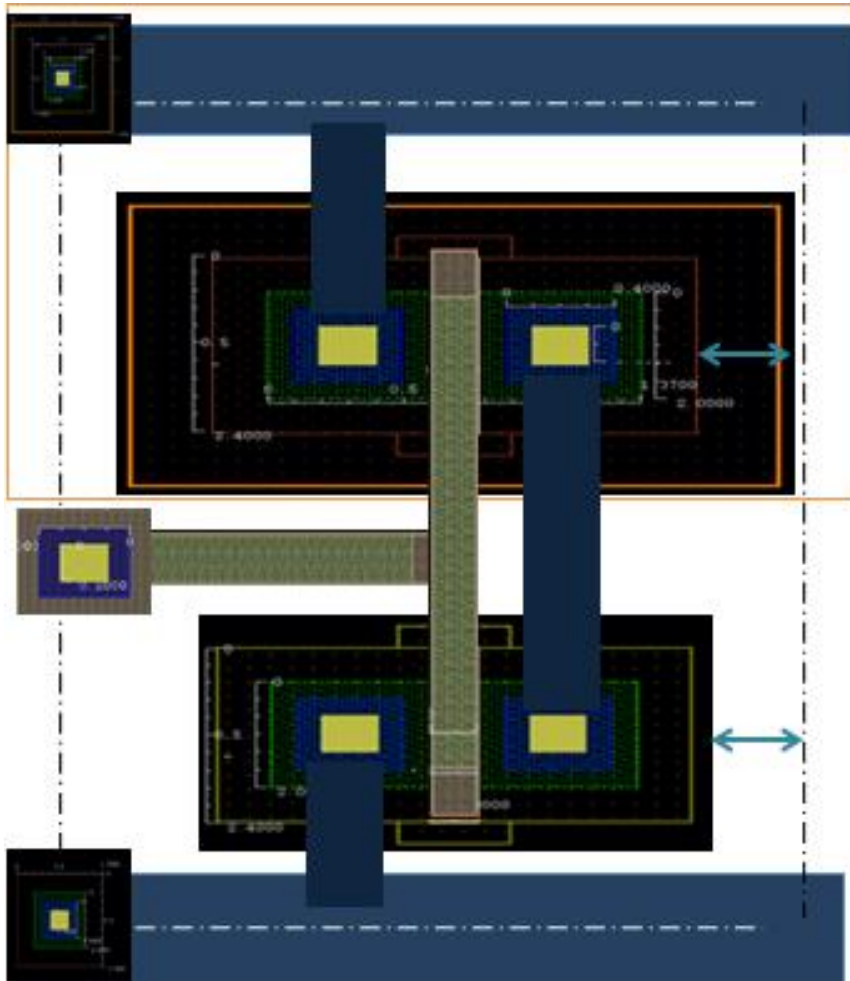
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Standard Cells



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Layout of CMOS Inverter



Summary for sonakshi/pseudonmos/av_extracted

instance count totals:

lib	cell	view	total
analogLib	pcapacitor	symbol	10
analogLib	presistor	symbol	10
gpd180	nmos	ivpcell	1
gpd180	pmos	ivpcell	1



THANK YOU

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